

Solvent Extraction

Principles

and Practice

Second Edition, Revised and Expanded

edited by
Jan Rydberg
Michael Cox
Claude Musikas
Gregory R. Choppin



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*Second Edition, Revised and Expanded
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Preface

The partition of a solute between two immiscible solvents is a major technique in separation, both in the laboratory and in industry. This technique has been designated *liquid-liquid distribution* by the International Union of Pure and Applied Chemistry (IUPAC), but it is more commonly called *solvent extraction*, the term we use. Solvent extraction has been developed in a broad field of applications: (1) for studying inorganic and organic complex equilibria, either for fundamental understanding of solution equilibria and kinetics or for developing selective separation schemes; (2) for separations in analytical chemistry; (3) for large-scale industrial separation processes in the inorganic, organic, pharmaceutical, and biochemical industries; and (4) for industrial waste treatment. It also serves as a base for other analytical techniques, such as liquid partition chromatography and ion-selective electrodes.

The many uses of solvent extraction make the subject important for university students of chemistry and chemical technology. Some universities offer special courses in solvent extraction, whereas others include it as a minor topic in more comprehensive courses. Pilot-scale experiments on solvent extraction are common in chemical engineering curricula. Because of the breadth of the subject, the treatment in such courses is often scanty, and a satisfactory text is difficult to find in a form suitable for use directly with students.

To meet this demand, we felt it desirable to develop a simple text suitable not only for students but also for chemists in the field. However, no single scientist or engineer can be an expert in all parts of the field. Therefore, it seemed best to develop the book as a joint project among many expert authors, each of whom has decades of experience in research, teaching, or industrial development. The product is a truly international book at a high scientific and technical level. The task of making it homogeneous, consistent, and pedagogical has been undertaken by the editors.

The book is directed to third- to fourth-year undergraduate and post-graduate chemistry and chemical engineering students, as well as to researchers and developers in the chemical industry. The book is also intended for chemical engineers in industry who either have not kept up with modern developments or are considering use of this technique, as well as engineers who already are using this technique but desire to understand it better. Furthermore, the book should be useful to researchers in solvent extraction who wish to learn about its applications in areas other than their own.

The first edition of this book was published more than 10 years ago. Since then four large international conferences have summarized the latest developments in the field of solvent extraction. We have tried to incorporate the latest achievements of a fundamental type in this text, for instance, new types of solvent matrices and industrial applications, without burdening the text with new organic extractants or solvent combinations, which appear almost daily in the specialist literature. As far as possible, we have also rationalized the earlier text, concentrating on or removing outdated information. In doing

so we have added new contributors. We therefore hope that this text will be met with the same enthusiasm as the first edition.

After an introduction (Chapter 1), the following five chapters (Chapters 2–6) present the physical principles and formal expressions used in solvent extraction. They are followed by eight chapters (Chapters 7–14) of various industrial applications and two concluding chapters (Chapters 15 and 16) indicating the research frontiers and future developments in technology.

We, the editors, want to express our gratitude to the contributors, who made this book possible through their helpful suggestions and extensive efforts. Prof. Manuel Aguilar, Prof. Eckhart Blass, Prof. José Luis Cortina, Dr. Pier Danesi, Prof. Ingmar Grenthe, Prof. Jan-Olov Liljenzin, Dr. Marian Czerwiński, Prof. Philip Lloyd, Prof. Yizhak Marcus, Prof. Jerzy Narbutt, Dr. Susana Pérez de Ortiz, Dr. Hans Reinhardt, Dr. Gordon Ritcey, Prof. Ana María Sastre, Dr. Wallace Schulz, the late Prof. Tatsuya Sekine, Prof. David Stuckey, Dr. Hans Wanner, and Prof. Ronald Wennersten have worked hard and successfully on their own chapters and have given us much valuable guidance.

Jan Rydberg
Michael Cox
Claude Musikas
Gregory R. Choppin

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Solvent Extraction Principles and Practice

Introduction to Solvent Extraction

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1.1 WHAT IS SOLVENT EXTRACTION?

The term *solvent extraction*[†] refers to the distribution of a solute between two immiscible liquid phases in contact with each other, i.e., a two-phase distribution of a solute. It can be described as a technique, resting on a strong scientific foundation. Scientists and engineers are concerned with the extent and dynamics of the distribution of different solutes—organic or inorganic—and its use scientifically and industrially for separation of solute mixtures.

The principle of solvent extraction is illustrated in Fig. 1.1. The vessel (a separatory funnel) contains two layers of liquids, one that is generally water (S_{aq}) and the other generally an organic solvent (S_{org}). In the example shown, the organic solvent is lighter (i.e., has a lower density) than water, but the opposite situation is also possible. The solute A, which initially is dissolved in only one of the two liquids, eventually distributes between the two phases. When this distribution reaches equilibrium, the solute is at concentration $[A]_{\text{aq}}$ in the aqueous layer and at concentration $[A]_{\text{org}}$ in the organic layer. The *distribution ratio* of the solute

$$D = [A]_{\text{org}} / [A]_{\text{aq}} \quad (1.1)$$

*Retired.

[†]The International Union of Pure and Applied Chemistry (IUPAC) recommends the use of the term *liquid-liquid distribution*. However, more traditionally the term *solvent extraction* (sometimes abbreviated SX) is used in this book.

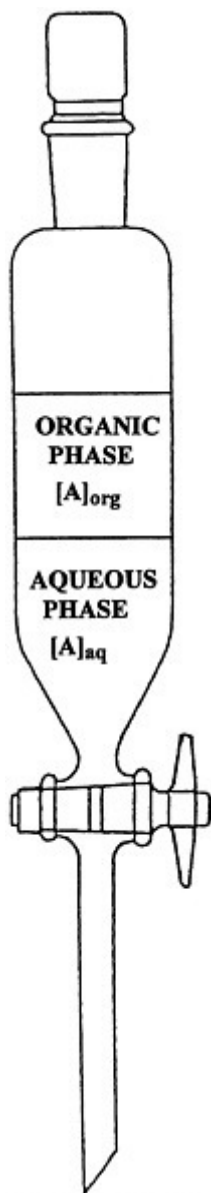


Fig. 1.1 A schematic representation of solvent extraction (liquid-liquid distribution). A solute A is distributed between the upper layer, for example an organic solvent, and the lower layer, an aqueous phase.

is defined as the ratio of “the total analytical concentration of the substance in the organic phase to its total analytical concentration in the aqueous phase, usually measured at

equilibrium” [1], irrespective of whether the organic phase is the lighter or heavier one. If a second solute B is present, the distribution ratio for the various solutes are indicated by D_A , D_B , etc. If D_B is different from D_A , A and B can be separated from each other by (single or multistage) solvent extraction. D is also called the *distribution coefficient* or *distribution factor*; we here prefer the expression *distribution ratio*.

For practical purposes, as in industrial applications, it is often more popular to use the *percentage extraction %E* (sometimes named the *extraction factor*), which is given by

$$\%E = 100D/(1+D) \quad (1.2)$$

where D is the distribution ratio of the solute (or desired component). For $D = 1$, the solute is evenly distributed between the two phases. A requirement for practical use of solvent extraction is that a reasonable fraction (percentage) of the desired component is extracted in a single operation (or *stage*).

Solvent extraction is used in numerous chemical industries to produce pure chemical compounds ranging from pharmaceuticals and biomedical to heavy organics and metals, in analytical chemistry and in environmental waste purification. The scientific explanation of the distribution ratios observed is based on the fundamental physical chemistry of solute-solvent interaction, activity factors of the solutes in the pure phases, aqueous complexation, and complex-adduct interactions. Most university training provides only elementary knowledge about these fields, which is unsatisfactory from a fundamental chemical standpoint, as well as for industrial development and for protection of environmental systems. Solvent extraction uses are important in organic, inorganic, and physical chemistry, and in chemical engineering, theoretical as well as practical; in this book we try to cover most of these important fields.

None of the authors of this book is an expert in all the aspects of solvent extraction, nor do we believe that any of our readers will try to become one. This book is, therefore, written by authors from various disciplines of chemistry and by chemical engineers. The “scientific level” of the text only requires basic chemistry training, but not on a Ph.D. level, though the text may be quite useful for extra reading even at that level. The text is divided in two parts. The first part covers the fundamental chemistry of the solvent extraction process and the second part the techniques for its use in industry with a large number of applications. In this introductory chapter we try to put solvent extraction in its chemical context, historical as well as modern. The last two chapters describe the most recent applications and theoretical developments.

1.2 SOLVENT EXTRACTION IS A FUNDAMENTAL SEPARATION PROCESS

Under normal conditions, matter can appear in three forms of aggregation: solid, liquid, and gas. These forms or *physical states* are consequences of various interactions between the atomic or molecular species. The interactions are governed by *internal chemical properties* (various types of bonding) and *external physical properties* (temperature and

pressure). Most small molecules can be transformed between these states (e.g., H_2O into ice, water, and steam) by a moderate change of temperature and/or pressure. Between these physical states—or *phases*—there is a sharp boundary (*phase boundary*), which makes it possible to separate the phases—for example, ice may be removed from water by filtration. The most fundamental of chemical properties is the ability to undergo such phase transformations, the use of which allows the simplest method for isolation of pure compounds from natural materials.

In a gas mixture such as the earth's atmosphere, the ratio of oxygen to nitrogen decreases slightly with atmospheric height because of the greater gravitational attraction of oxygen. However, the gravitational field of the earth is not enough for efficient separation of these gases, which, however, can be separated by ultracentrifugation and by diffusion techniques. In crushed iron ore it is possible to separate the magnetite crystals Fe_3O_4 from the silicate gangue material by physical selection under a microscope or by a magnetic field. In chemical engineering such separation techniques are referred to as *nonequilibrium processes*. Other common nonequilibrium processes are electrolysis, electrophoresis, and filtration.

In contrast to these we have the *equilibrium processes* of sublimation, absorption, dissolution, precipitation, evaporation, and condensation, through which the physical states of solid, liquid, and gas are connected. For example, the common crystallization of salts from sea water involves all three phases. Distillation, which is essential for producing organic solvents, is a two-step evaporation (liquid $\Rightarrow$ gas) condensation (gas $\Rightarrow$ liquid) process.

In Fig. 1.2, phase transformations are put into their context of physical processes used for separation of mixtures of chemical compounds. However, the figure has been drawn asymmetrically in that two liquids (I and II) are indicated. Most people are familiar with several organic liquids, like kerosene, ether, benzene, etc., that are only partially miscible with water. This lack of miscibility allows an equilibrium between two liquids that are separated from each other by a common phase boundary. Thus the conventional physical system of three phases (gas, liquid, and solid, counting all solid phases as one), which ordinarily are available to all chemists, is expanded to four phases when two immiscible liquids are involved. This can be of great advantage, as will be seen when reading this book.

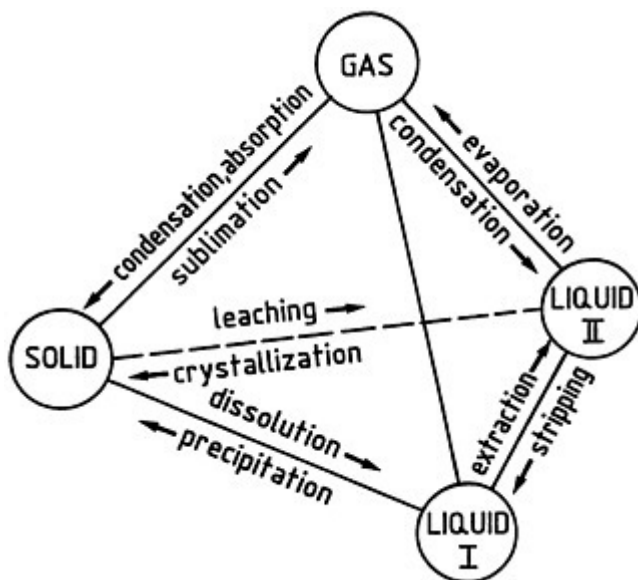


Fig. 1.2 Model of a four-phase system consisting of two liquid phases (e.g., an aqueous and an organic phase) in equilibrium with a gas phase and a solid phase.

Solutes have differing solubilities in different liquids due to variations in the strength of the interaction of solute molecules with those of the solvent. Thus, in a system of two immiscible or only partially miscible solvents, different solutes become unevenly distributed between the two solvent phases, and as noted earlier, this is the basis for the *solvent extraction technique*. In this context, “solvent” almost invariably means “organic solvent.” This uneven distribution is illustrated in Fig. 1.3, which shows the extractability into a kerosene solution of the different metals that appear when stainless steel is dissolved in aqueous acid chloride solution. The metals Mo, Zn, and Fe(III) are easily extracted into the organic solvent mixture at low chloride ion concentration, and Cu, Co, Fe(II), and Mn at intermediate concentration, while even at the highest chloride concentration in the system, Ni and Cr are poorly extracted. This is used industrially for separating the metals in super-alloy scrap in order to recover the most valuable ones.

The three main separation processes between solid, gas, and liquid have long been known, while solvent extraction is a relatively new separation technique, as is described in the brief historical review in next two sections. Nevertheless, because all solutes (organic as well as inorganic) can be made more or less soluble in aqueous and organic phases, the number of applications of solvent extraction is almost limitless. Since large-scale industrial solvent extraction is a continuous process (in contrast to laboratory batch processes) and can be

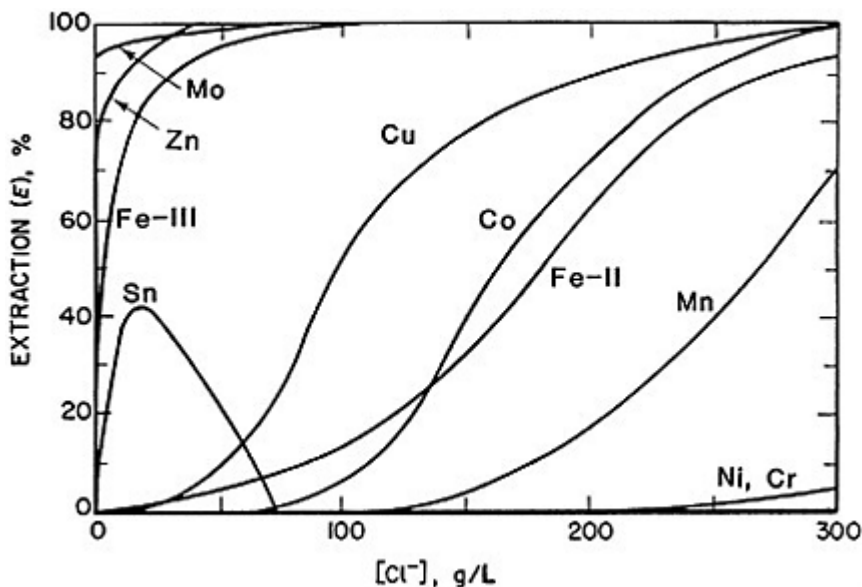


Fig. 1.3 Percentage of extraction of various metals from a solution of dissolved stainless steel scrap, at 40°C. The organic phase is 25% tertiary amine (Alamine 336), 15% dodecanol (Loral C12) and 60% kerosene (Nysolvin 75A). The aqueous phase is a CaCl_2 solution at pH 2.

made more selective than the conventional gas-liquid-solid separation techniques, it offers numerous industrial possibilities to achieve desired separation efficiently and economically.

1.3 EARLY STEPS TOWARD THE USE OF SOLVENT EXTRACTION

Around 500 B.C.E. the Greek philosophers recognized four elements: earth, water, phlogiston (~air), and fire. This view harmonizes with the present concept of three physical states of aggregation (solid, liquid, gas) and heat. Aristotle (~350 B.C.E.) emphasized that these “elements” were not eternal, but could be changed into each other. Five thousand years earlier, scientists already had found that when certain green minerals were heated in a coal fire, metallic copper was obtained. In Aristotle’s time it was known how to produce metals such as copper, gold, tin, lead, silver, iron, mercury, and arsenic. Even earlier, by transmuting certain “earths” with fire, ceramics and glasses had been produced. Many of these arts were probably developed by the Egyptians, the first true chemists. The word “*alchemy*” is derived from Arabic and Greek and is supposed to mean “art of transmutations as practiced by the Egyptians.” Fruit juices were fermented, oils and fats were squeezed out of vegetables and animal parts, and purified by digestion with earths, bones, etc. Crucibles, retorts, and even distillation equipment seem to have

been in use; see Fig. 1.4. We must think of these early alchemists as endlessly mixing, heating, boiling, digesting, cooling, etc., everything they could collect from nature. The purpose of these transmutations varied: for lamps, weapons, pigments, perfumes, and poisons; for cosmetics and medicines to prevent aging and to prolong life (*elixir vitae*); for tanning chemicals, soap, anesthetics, and also for making gold. In fact, they did succeed in producing gold-like metals (e.g., brass). For example, it is known that they heated odorous leaves in alkaline water with fats and oils, so that ointments and perfumes could be enriched in the cooled solidified fat, and that these products were extensively used in the ancient courts, and perhaps even among the general population. If this is considered to be solvent extraction, it is truly one of the oldest chemical techniques. It is also likely that the Egyptians knew how to distill alcohol, long before it is described by the Arab Kautilya and the Greek Aristotle about 300 B.C.E. (See also Ref. [2].)

This experimentation more or less came to a halt during the Greek civilization. The Greeks were philosophers, and not so much experimentalists; Aristotle was a philosopher and a systematizer (systems technician, in modern language), not an experimentalist. The Greeks were followed by the Romans who were administrators, and by the Christians who considered alchemy to be ungodly. Although alchemy was practiced during subsequent centuries, particularly in the Arabian world, it became suspect and was banned by many rulers (though encouraged in secrecy by others). About 500 years ago, alchemy was rather openly revived in Europe, particularly at local courts, and progressed within a few centuries into modern science.

Digestion of various earths (or digested earths) with alcohol produces many organic solvents (ether, acetone, etc.). These solvents could be obtained in pure form through distillation. Such organic solvents could have been produced many thousands years ago, because of the obvious knowledge of distillation [2]. However, 200 years ago only a few pure solvents seem to be known: besides the natural water and oils (and kerosene) only alcohol, ether, and “ethereal oils” were acknowledged. It is difficult to trace organic solvents far back in history. The reason may simply be that organic compounds obtained by distillation of mixtures of natural products were found to be rather uninteresting (except for alcohol), because at that time they seemed to have very little practical value, and they certainly could not be used to produce gold. Because solvent extraction requires pure organic solvents of limited aqueous miscibility, it is then understandable why solvent extraction historically is considered (perhaps falsely) to be a newcomer among chemical separation methods.



Fig. 1.4 (a) Equipment used by alchemists, according to an Alexandrian manuscript (about 300 B.C.E. to A.D. 300). (b) Apparatus for fractionated distillation. Front page of *Philosophi ac Alchimistae Maximi* by Johannes Greininger, Strasbourg, 1531. The original work is ascribed to the great eighth-century Arab alchemist, Abu Musa Jabir.

1.4 SOLVENT EXTRACTION IN THE DEVELOPMENT OF MODERN CHEMISTRY

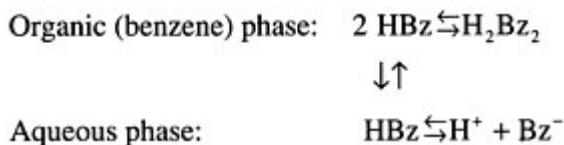
The early nineteenth century saw an enormous development in chemistry. Not only were numerous inorganic and organic compounds synthesized, they were also characterized through skillful analyses. Pure compounds, absolutely essential for the analyst, were identified from measurements of weights and volumes, melting and boiling points, densities, refraction, and crystal structure. It was then found that many inorganic salts, such as chlorides of iron, mercury, and gold, dissolved in and could be recrystallized from alcohols and ethers.

1.4.1 The Distribution Ratio and the Extraction Constant

Inorganic extractions seem to have come into practical use without any great notice. Although Peligot in 1842 reported that uranyl nitrate could be recrystallized from ether, he never mentioned extraction of this salt from aqueous solutions. In textbooks after 1870, however, it is stated briefly that “ether can even withdraw sublimate (HgCl_2) from aqueous solution.” It was also reported, for example, that cobalt thiocyanate is weakly extracted by ether, better by amyl alcohol, and even better by a mixture of both.

The practical use of solvent extraction for separation and purification of different substances led Berthelot and Jungfleisch to investigate the distribution of a large number of organic and inorganic compounds between ether or carbon disulfide and water. In 1872, they introduced the term *distribution factor* (“coefficient de partage”) [Eq. (1.1)] to describe how the distribution of a solute depended on its concentration in the organic and aqueous phases. However, the regularities observed were not general.

About 20 years later, Nernst realized that it was necessary to take into account the different reactions of the solute in each phase, such as dimerization in the organic phase and dissociation in the aqueous phase. Thus the distribution of benzoic acid (HBz) between the organic (benzene) phase and water could be written



In 1891, Nernst realized that only if the solute has the same molecular weight in the organic phase as in the aqueous phase, the distribution ratio would be independent of the concentration of the solute, or *distribuend*. He proposed the simple relation:

$$K_D = [\text{HBz}]_{\text{org}} / [\text{HBz}]_{\text{aq}} \quad (1.3)$$

and demonstrated the validity of this principle for the distribution of a large number of

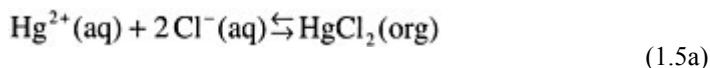
organic and inorganic compounds between organic solvents and water (*the Nernst distribution law*). K_D is referred to as the *distribution constant*. [Note that K_D is sometimes termed *partition coefficient* (or *constant*), and then abbreviated P .]

The primary parameter in solvent extraction is *the measured distribution ratio*, where it is up to the writer to define what is being measured, indicating this by an appropriate index. In the Nernst distribution experiment described earlier, the analytically measured concentration of benzoic acid is in the aqueous phase $[Bz]_{aq,tot} = [HBz]_{aq} + [Bz^-]_{aq}$, and in the organic phase $[Bz]_{org,tot} = [HBz]_{org} + [H_2Bz_2]_{org}$. Thus the measured distribution ratio, abbreviated D_{Bz} , becomes

$$D_{Bz} = \frac{[Bz]_{org,tot}}{[Bz]_{aq,tot}} = \frac{[HBz]_{org} + 2[H_2Bz_2]_{org}}{[HBz]_{aq} + [Bz^-]_{aq}} \quad (1.4)$$

From this it follows that the analytically measured distribution ratio is a constant only in systems that contain a single molecular species of the solute (or desired component).

Such considerations were extended to metal complexes in 1902 by Morse, who studied the distribution of divalent mercury between toluene and water at various Hg^{2+} and Cl^- concentrations. By taking complex formation in the aqueous phase into consideration Morse could determine the formation constants of $HgCl^+$ and $HgCl_2$ from distribution measurements, as well as the distribution constant of the neutral complex $HgCl_2$. The overall *extraction reaction* can be written



for which one defines an *extraction constant*, K_{ex}

$$K_{ex} = \frac{[HgCl_2]_{org}}{[Hg^{2+}]_{aq}[Cl^-]_{aq}^2} \quad (1.5b)$$

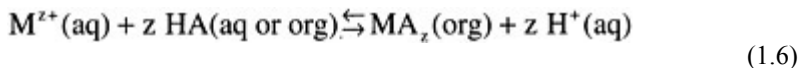
Although K_{ex} neither describes the intermediate reaction steps (see Chapters 3 and 4) or kinetics of the reaction (Chapter 5), nor explains why or to what extent $HgCl_2$ dissolves in the organic solvent (Chapter 2), the extraction reaction is a useful concept in applied solvent extraction, and K_{ex} values are commonly tabulated in reference works [3–4].

1.4.2 Extractants

During the years 1900 to 1940, solvent extraction was mainly used by the organic chemists for separating organic substances. Since in these systems, the solute (or desired component) often exists in only one single molecular form, such systems are referred to as *nonreactive extraction* systems; here the distribution ratio equals the distribution constant.

However, it was also discovered that many organic substances, mainly weak acids, could complex metals in the aqueous phase to form a complex soluble in organic

solvents. A typical reaction can be written



which indicates that the organic acid HA may be taken from the aqueous or the organic phase. This is an example of *reactive extraction*. It became a tool for the analytical chemist, when the extracted metal complex showed a specific color that could be identified spectrometrically. The reagent responsible for forming the extractable complex is termed the *extractant*.

The industrial use of solvent extraction of inorganic compounds grew out of the analytical work. As both areas, analytical as well as industrial, needed both better extractants and an understanding of the reaction steps in the solutions in order to optimize the applications, theoretical interpretations of the molecular reactions in the solutions became a necessity, as will be described in later chapters.

The increased use of computer graphics for modeling molecular structures and chemical reactions has opened a path for the synthesis of tailor-made extractants. Thus the future promises new varieties of extractants with highly selective properties for the desired process.

1.4.3 Industrial Use of Solvent Extraction

A surge in interest in solvent extraction occurred in the decades of the 1940s and 1950s initiated by its application for uranium production and for reprocessing of irradiated nuclear materials in the U.S. Manhattan Project. The first large-scale industrial solvent extraction plant for metals purification was built in 1942 by Mallinckrodt Chemical Co., St. Louis, for the production of ton amounts of uranium by selective extraction of uranyl nitrate by ether from aqueous solutions. The high degree of purity (>99.9%) required for use of uranium in nuclear reactors was achieved. An explosion led to the replacement of the ether by other solvents (dibutylmethanol and methylisobutylketone). At the same time new types of more efficient metal extractants were introduced, e.g., tri-*n*-butylphosphate in 1945 and trioctylamine in 1948. This activity became a great stimulus to the non-nuclear industry, and solvent extraction was introduced as a separation and purification process in a large number of chemical and metallurgical industries in the 1950s and early 1960s. For example, by leaching copper ore with sulphuric acid followed by extraction of this solution with an organic hydroxyaryloxime dissolved in kerosene, several million tons of copper (30% of world production) is now produced annually. This and many other processes are described in later chapters.

For these applications, the technique of solvent extraction had to be further developed and with this a new terminology was also developed. This can be illustrated by considering a process where a desired component in an aqueous solution is extracted with an organic reagent (extractant) dissolved in another organic liquid; note here that the term “organic solvent” is not used because of possible confusion. The term “solvent” could be used for the whole organic phase or for the organic liquid in which the organic extractant is dissolved. Thus the term generally given to the latter is (*organic*) *diluent*.

While in laboratory experiments the extraction vessel may be a test tube, or more conveniently some kind of separation funnel (Fig. 1.1), this is not suited for industrial use. Industry prefers to use continuous processes. The simplest separation unit is then the mixer-settler, or some clever development of the same basic principle, as described in Chapter 9. Figure 1.5 pictures a simple mixer-settler unit, here used for the removal of iron from an acid solution also containing nickel and cobalt (the same systems as in Fig. 1.3). The mixer (or *contactor*) is here simply a vessel with a revolving paddle that produces small droplets of one of the liquid phases in the other. This physical mixture flows into and slowly through the separation vessel, which may be a long tank; through the influence of gravity the two phases separate, so that the upper organic kerosene-octanol-amine phase contains the Fe(III) and the lower aqueous CaCl_2 phase contains the Co(II) and Ni(II). Numerous variations of the construction of mixer-settlers (or MS-units, as they are abbreviated) exist (Chapter 9), often several joined together into MS-batteries. Many such will be described later on.

A diagram of a full basic process is given in Fig. 1.6 to illustrate the common terminology. The incoming aqueous solution is called the *feed*. It is

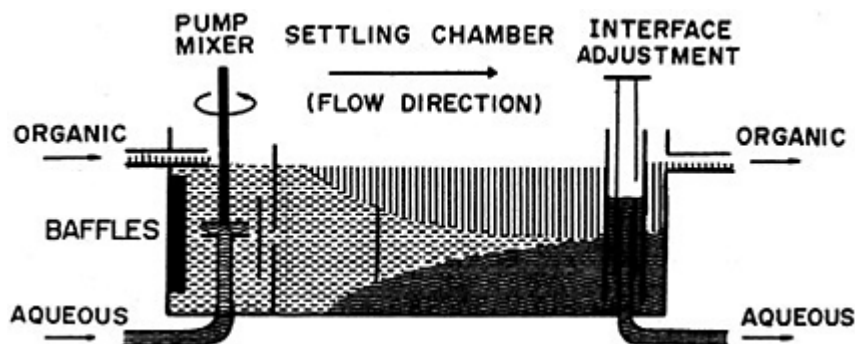


Fig. 1.5 The principle of a mixer-settler unit, e.g., for separation of iron(III) from nickel and cobalt.

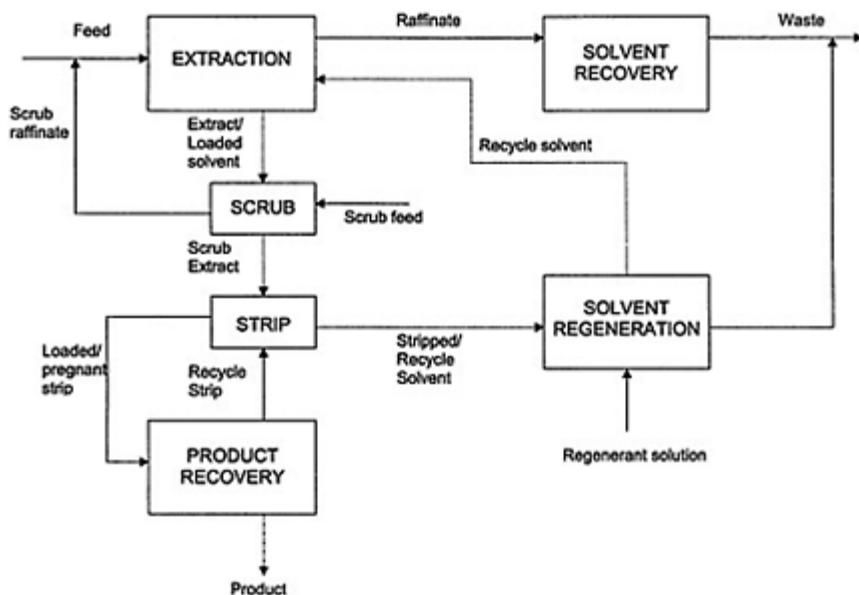


Fig. 1.6 Typical flowsheet of a solvent extraction circuit.

contacted with the (*recycled*) *solvent phase* in a mixer-settler unit. Here we do not indicate the exact type of unit, but only its function (extraction), as commonly is done. After extraction and separation of the phases, the depleted phase becomes the *raffinate* and the enriched solvent phase becomes the *extract* or *loaded (or pregnant) solvent*. The raffinate may undergo a solvent recovery stage to remove any entrained solvent before exiting the process. The extraction process is rarely specific so that other solutes may be co-extracted with the main component. These impurities may be removed with an aqueous *scrub solution* in a scrub stage producing a scrub extract and a scrub raffinate containing the impurities. The latter may return to the feed solution to maintain an overall water balance. The scrubbed extract is now contacted with another aqueous solution to strip or back-extract the desired component. The stripped solvent then may undergo some regeneration process to prepare the solvent phase for recycle. The *loaded (pregnant) strip solution* then is treated to remove the desired product and the strip solution is recycled. One of the important aspects of this flowsheet is that, wherever possible, liquid phases are recovered and recycled. This is important from both an economic and an environmental standpoint. Figure 1.6 shows a situation where each process—extraction, scrubbing, and stripping—occurs in a single operation or stage. This is generally not efficient because of the finite value of the distribution coefficient. Thus if D has a value of 100 (i.e., $E=99\%$), then after one extraction, the organic extract phase will contain approximately 99 parts and the aqueous phase 1 part. To achieve a greater extraction, the aqueous raffinate should be contacted with another portion of the solvent after which the new organic phase contains 0.99 parts, and the aqueous raffinate now only 0.01 part. Therefore, two extraction stages will provide 99.99% extraction (with “2 volumes” of the

organic phase, but only “1 volume” of the aqueous phase).

Three different ways of connecting such stages are possible: namely, *co-current*, *cross-current*, and *counter-current* (see Fig. 1.7). In co-current extraction, the two phases flow in the same direction between the various contactors. A simple inspection of the diagram will show that with this configuration no advantage is gained over a single contact because, providing equilibrium is reached in the first contactor, the separated flows are in equilibrium when entering the second contactor so no change in relative concentrations will occur. In the second configuration (b), cross-current, the raffinate is contacted with a sample of fresh solvent. This is the classical way of extracting a product in the laboratory when using a separatory funnel and will give an enhanced recovery of the solute. However, on an industrial scale, this is seldom used because it

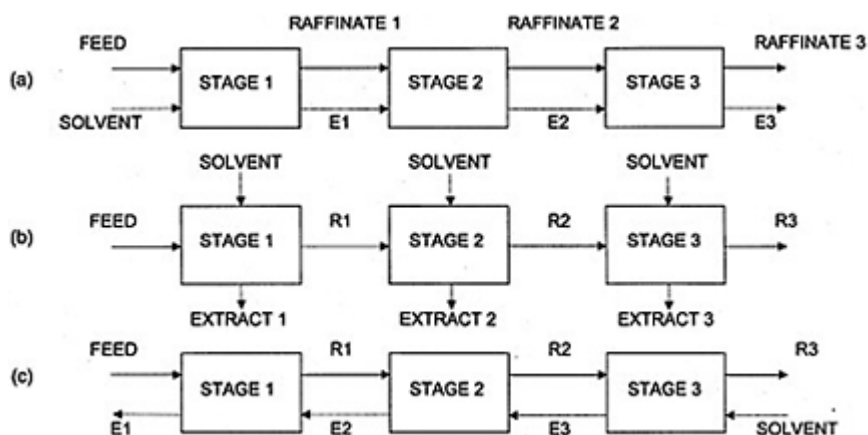


Fig. 1.7 Arrangements of solvent extraction stages: (a) co-current; (b) cross-current; and (c) counter-current.

results in the production of a multitude of product phases containing a reducing concentration of the desired solute. These have to be combined before stripping resulting in a much larger volume of loaded solvent to be treated with consequences for plant size and economics. The third configuration (c), counter-current, is the one generally chosen by industry. The phase volumes remain constant and by feeding the two phases, feed and solvent, at opposite ends of the bank of contactors, the driving force for extraction, i.e., the solute concentration difference between the two phases, is maximized. Chapters in the second part of the book will extend this discussion.

The need to use multiple extraction to achieve efficient extraction required the development of new types of continuously working extractors, especially mixer-settlers and pulsed columns, which were suitable for remotely controlled operations. These new extractors could be built for continuous flow and in multiple stages, allowing very efficient isolation of substances in high yield. A good example is the production of rare earth elements in >99.999% purity in ton amounts by mixer-settler batteries containing hundreds of stages. These topics will be further developed in Chapters 6 and 7.

In the early analytical applications of solvent extraction, optimal extraction or

separation conditions were obtained empirically. This was unsatisfactory and general mathematical descriptions were developed by a number of researchers in many countries. This was especially important for large-scale industrial use and is an activity that continues today almost entirely with computers.

1.5 MODERN TRENDS IN APPLICATIONS OF SOLVENT EXTRACTION

What is the future of solvent extraction? We will try to indicate some areas in which we expect important advances in the future. Some are more theoretical, others are applied, but all are related to other areas of chemistry or to other fields of science and technology. The simple answers given here do not require the reading of the subsequent chapters, but do require some good fundamental chemical knowledge.

1.5.1 *Solution Theories*

A 10% change in the distribution ratio, which can be measured easily and accurately in the distribution range 0.01–100 (the one most commonly used), corresponds to an energy change of only 0.2 kJ/mole. Thus distribution ratio measurements offer a method to investigate low energy reactions in solutions, such as weak solute-solute and solute-solvent interactions, especially in the organic phase. So far, this technique has been only slightly exploited for this purpose. It is particularly noteworthy to find how few thermodynamic studies have been made of the distribution or extraction constants, as enthalpy and entropy values give a good indication of the driving force of the extraction and also may indicate the structure of the molecular species in the organic solvent.

Separation of metals by solvent extraction is usually based on the various complexing properties of the metals (Chapter 3). Separation systems may be chosen on the basis of complexity constants obtained from the literature. However, the literature often shows different values for “same systems” causing considerable concern for process design chemists. There is an obvious need for an objective presentation of the uncertainty in the published equilibrium constants, however conditional they may be.

Although theories of solution (Chapter 2) and formation of extractable complexes (Chapters 3 and 4) now are well advanced, predictions of distribution ratios are mainly done by comparison with known similar systems (Chapter 3). Solvatochromic parameters, solubility parameters, and donor numbers, as discussed in Chapters 2–4, are so far mainly empirical factors. Continuous efforts are made to predict such numbers, often resulting in good values for systems within limited ranges of conditions. It is likely that these efforts will successively encompass greater ranges of conditions for more systems, but much still has to be done. In the future, theory may allow the assignment of exact numbers to the solvatochromic parameters, thus also permitting theoretical predictions of distribution constants for known as well as for hypothetical (not yet synthesized) extractants.

Chemical quantum mechanics (Chapter 16) are now promising to contribute to prediction of distribution ratios, particularly when used in interactive computer modeling

of chemical structures and reactions. This will provide better understanding of solute-solvent interactions and the theory of solubility, which is the foundation for prediction of distribution ratios. Progress is likely to speed up as efficient computing programs with large data bases become more easily available on the market at more attainable prices. Such interactive computer research will be able to explain poor extractability due to steric hindrance, hydration, the synergistic effect of specific adducts, etc., and, as a consequence, give clues to better extraction reagents and conditions.

A combination of known extractants can increase the net extraction both in degree and kinetics through synergistic extraction. This subject also is poorly explored, though it is of considerable importance to industrial solvent extraction processes.

1.5.2 Kinetics

All chemical reactions occur with a certain rate. For aqueous systems, in which no redox reactions occur (e.g., simple complex formation), equilibrium is often attained rapidly, which is also true for adduct reactions in the organic phase, all at normal temperatures. However, the transfer of a solute from an aqueous solution to an organic solvent occurs via a phase boundary; such reactions depend on several parameters and may be quite slow. Therefore, the kinetics of solvent extraction, which is of paramount importance to all industrial applications, is largely determined by the interfacial chemistry. To speed up phase transfer, the engineer tries to maximize the interfacial surface by, e.g., violent stirring, producing billions of small droplets. Though the theory is well advanced (see Chapters 5 and 9), the interface is very complicated and its properties are difficult to investigate, particularly at the molecular level. Few applicable techniques are available, and the results are often difficult to interpret. The coalescence of the droplets (to produce two clean phases that are easy to separate) presents another interfacial surface problem, and so-called phase reversals may further complicate the separation. This is particularly important in the development of new types of contactor. Much progress is required here and as such presents a challenge to future scientists and engineers.

The largest research effort in extraction kinetics is likely to be in the development of solvent extraction related techniques, such as various versions of liquid chromatography, liquid membranes, etc. These techniques require a detailed knowledge of the kinetics of the system to predict the degree of separation.

It is a practical fact that most industrial solvent extractions are carried out under nonequilibrium conditions, however close the approach may be; for example, centrifugal contactor-separators (Chapter 9) rarely operate at distribution equilibrium. An interesting possibility is to expand this into extractions further from equilibrium, if the kinetics of the desired and undesired products are different. Such operations offer a real technological challenge.

1.5.3 Equipment and Processes

Laboratory solvent extraction studies are often carried out with test tubes, a process that is cheaper but more laborious, time consuming, and usually yields more scattered data, than using semicontinuously stirred baffled beakers or Lewis-type cells. Highly precise

distribution data can be obtained efficiently either “mechanically” by continuous flow centrifugal separators, or “physically” by using special phase separating filter devices; some of these methods are described in Chapter 4. Although these techniques are going to be improved, new, more advanced methods are not likely to appear in the near future.

Specific reagents with exotic structures are costly to produce, which requires the experimentalist to use as small amounts as possible of the reagents. This leads to a desire to shift from “milli-experiments” (e.g., mixing 10 mL of each phase) to micro experiments (mixing 50 μ L of each phase), which is now taking place, to even smaller sizes (volumes and amounts)—the *nano experiments*. Progress in this field requires a high degree of ingenuity.

Although single-stage laboratory techniques provide the first step toward multistage industrial processes, such process development usually requires small-scale multistage and pilot-plant scale equipment. A large number of excellent designs are available, and we consider further fundamental improvements unlikely.

The industrial application of solvent extraction is a mature technique, and it is now possible to move from laboratory experiments on a new extraction system to full industrial practice with little technological risk. There is a sufficient variety of large-scale equipment available to cope with most problems encountered in application, although much of the equipment remains rather massive. Attempts to miniaturize, for instance, by using centrifugal forces to mix and separate phases, still has to be developed further.

Many industrial processes begin with a leaching step, yielding a slurry that must be clarified before solvent extraction. The solid-liquid separation is a costly step. The solvent extraction of unclarified liquids (“solvent-in-pulp”) has been proposed to eliminate solid-liquid separation. The increased revenue and reduced energy cost make this an attractive process, but many problems remain to be solved: loss of metals and extractants to the solid phase, optimization of equipment design, effluent disposal, etc.

An essential step in industrial solvent extraction is the regeneration of the extractant. This can be done in many ways, e.g., by distillation, evaporation, or stripping (back-extraction). While distillation and evaporation do not discriminate between solutes (the diluent is simply removed by heating), stripping, by careful choice of strip solution and conditions, can be made highly selective. Alternatively, all the solutes can be stripped and then subjected to a selective extraction by changing the extractant; examples of both types of process will be found in Chapter 13. The possibilities are many, and it may be worthwhile to explore new paths.

Membrane extraction is a relatively new technique for solvent extraction, in which a solute is transferred from one aqueous phase to another through a membrane holding an extractant dissolved in a diluent. This ingenious scheme has been only slightly explored, though it offers great potential for the future, e.g., for waste water cleaning.

The step from laboratory experiments via pilot plants to industrial scale requires serious consideration of all the points here; practical experience is invaluable in order to avoid mistakes and excess costs, as indicated in Chapter 7.

1.5.4 Organic Chemistry and Biochemistry

There is a continuing demand for selective extractants that can be developed, for

example, by tailor making organic molecules with pre-organized metal binding sites. The development of such extractants may be designed on computer based models. Environmental considerations require that the extractants and diluents used are either nontoxic or recovered within the process. The increasing combination of extraction and distillation, as used in biotechnology (Chapter 10), places a whole host of new demands on the diluents employed.

In the extraction of biologically active compounds, care must be taken to avoid the loss of activity that often occurs by contact with organic diluents. Thus a series of systems have been developed specifically with these compounds in mind. The first of these uses mixtures of aqueous solutions containing polymers and inorganic salts that will separate into two phases that are predominately water. A second system uses supercritical conditions in which the original two-phase system is transformed into one phase under special temperature-pressure conditions. Also the active organic compound can be shielded from the organic diluent by encapsulation within the aqueous center of a micelle of surface active compounds. All these systems are currently an active area for research as is discussed in Chapter 15.

1.5.5 Separation of Inorganic Compounds

Separation schemes have been developed for all the elements in the periodic system, both in ionic and molecular forms. Some of these substances are toxic, putting severe restraints on plant operation and the environment. For example, many solvents are bioactive, and increasingly stringent control is needed to prevent their spread in the biosphere. Though the techniques are advanced, demands for improved safety will persist and require new types of equipment or processes.

The production of electricity from nuclear fission energy is accompanied by formation of radioactive waste, of which the larger hazard is the presence of long-lived transuranium isotopes. The problems associated with this waste are still debated, but if the transuranium isotopes could be removed by “exhaustive” reprocessing and transmuted in special nuclear devices, the hazard of the waste would be drastically reduced (Chapter 12). This may require new selective extractants and diluents as well as new process schemes. Research in this field is very active.

The mineral industry, where ores produce pure metals, requires a number of process stages that can be carried out either through pyrometallurgical treatment or by the use of wet chemical separation processes (*hydrometallurgy*). The former produces off gases and slags, which may be environmentally hazardous. The latter introduces chemistry into the mineral industry, and is often met with strong criticism among traditional metallurgists. It is therefore up to the hydrometallurgical chemist to develop schemes for metal production from ores that are economically competitive and environmentally advantageous. Here, the field is rather open for new developments. A specifically interesting trait is the use of “in-situ” leaching of ore bodies, which circumvents conventional mining with its sometimes devastating environmental effects, e.g., in open-pit mining.

In ore processing, the essential minerals are enriched through flotation of the crushed ore, using an aqueous solution containing flotation reagents (we may label them *floatants*),

that selectively attach to the mineral surface and—with the aid of air bubbles—lift them up to the surface where they can be skimmed off, while the “gangue” material stays at the bottom. Alternatively, the flotants are made magnetic, allowing the use of magnetic separation as commonly used in the mineral industry. These flotants are chemically very similar to the chelating extractants used in solvent extraction processes. An interesting subject here is collaboration between these two mineral processing fields to produce new and more effective reagents and separation processes.

1.5.6 Environmental Aspects

The future will bring further increase in concern over the environmental impact of chemical operations. The liquid effluents must not only be controlled, they must also be rendered harmless to the environment. This requires removal of the hazardous substances. For many of the dilute waste solutions, solvent extraction has proved to be an effective process. This is even more true for recycling of mixed metals from various industries. Nevertheless, the increasing amounts of wastes from human activities require much more to be done in this field.

In principle, solvent extraction is an environmentally friendly process with no air or water pollution, provided the plant flows are properly designed. It could, therefore, replace many of the present polluting processes. A particular problem, however, is the solvent extraction effluents, which may contain biochemically active substances posing “new” hazards to the environment. These can be handled by various solid sorbents, which then can be incinerated, but the advantage of the solvent extraction process may be lost. There is, therefore, a demand for biodegradable and environmentally benign solvent phases. In the future, additional attention is required to this field.

1.5.7 Spin-off

The principle of solvent extraction—the distribution of chemical species between two immiscible liquid phases—has been applied to many areas of chemistry. A typical one is liquid partition chromatography, where the principle of solvent extraction provides the most efficient separation process available to organic chemistry today; its huge application has become a field (and an industry!) of its own. The design of ion selective electrodes is another application of the solvent extraction principle; it also has become an independent field. Both these applications are only briefly touched upon in the chapter of this book on analytical applications (Chapter 14), as we consider them outside the scope of this work. Nevertheless, fundamental research on solvent extraction will provide further important input into these two applications. It is likely that fundamental solvent extraction research will continue to contribute to the development of further selective analytical techniques.

The driving force of the transport of salts, proteins, etc., through the cell membrane from the nucleus to the body fluids, and vice versa, is a complicated biochemical process. As far as is known, this field has not been explored by traditional solution chemists, although a detailed analysis of these transfer processes indicates many similarities with solvent extraction processes (equilibrium as well as kinetics). It is possible that studies of

such simpler model systems could contribute to the understanding of the more complicated biochemical processes.

Solvent extraction deals with the transport of chemical substances from one phase into another one, the chemical kinetics of this process, and the final equilibrium distribution of the substances between the two phases. Such transport and distribution processes are the motors that make life in biological systems possible. Fundamental studies of such “solvent extraction” processes contribute to the better understanding of all processes in nature. Here, only the lack of imagination stands in the way of important new scientific discoveries.

1.5.8 Conclusion

Solvent extraction is a mature technique in that extensive experience has led to a good understanding of the fundamental chemical reactions. At the same time, compared to many other chemical separation processes like precipitation, distillation, or pyrometallurgical treatment, the large-scale application of solvent extraction is, nevertheless, a young technique. New reagents are continually being developed, spurred on by computer modeling, and more efficient contacting equipment is coming into use. Considering such factors as demands for higher product purity, less pollution, and the need for recovering substances from more complex matrices and more dilute resources, the efficiency and high selectivity of solvent extraction should make it an increasingly competitive separation process both in research and in industry.

1.6 HOW TO USE THIS BOOK

The first part of the book contains basic solution theory and thermodynamics (Chapter 2); a survey of the effects on the distribution ratio of changes in parameter values, such as concentration of metals, complex formers and other reactants, pH, temperature, etc., (Chapter 3); measurement techniques, data collection, evaluation, and interpretation (Chapter 4); and kinetics (Chapter 5). The ionic strength is an essential factor in all aqueous systems, and how to cope with it is treated in a fundamental way in Chapter 6. These chapters sometimes go into such depth that they may mainly interest only the theoretical chemist or professional engineer; in some chapters we offer some choice by placing the most detailed parts in smaller type. It is not necessary to read these chapters before the applied ones, and some can be read independently, before or after the other more applied chapters.

The second part deals with applications of solvent extraction in industry, and begins with a general chapter (Chapter 7) that involves both equipment, flowsheet development, economic factors, and environmental aspects. Chapter 8 is concerned with fundamental engineering concepts for multistage extraction. Chapter 9 describes contactor design. It is followed by the industrial extraction of organic and biochemical compounds for purification and pharmaceutical uses (Chapter 10), recovery of metals for industrial production (Chapter 11), applications in the nuclear fuel cycle (Chapter 12), and recycling or waste treatment (Chapter 14). Analytical applications are briefly summarized

in Chapter 13. The last chapters, Chapters 15 and 16, describe some newer developments in which the principle of solvent extraction has or may come into use, and theoretical developments.

Apart from the first six chapters, each chapter has been written more or less independently but, we hope, consistently.

1.7 SOLVENT EXTRACTION NOMENCLATURE

Because scientists and engineers in chemistry use different symbols, a single set valid for all chapters in this book is not feasible. Moreover, it is desirable that both groups be familiar with the symbols used within the subfields in order to enhance communication. The symbols in this book closely follow IUPAC recommendations (see reference list). In the Appendix of this book, lists are given of the most commonly used symbols, except for those in Chapter 9, which has its own list of symbols.

1.8 LITERATURE ON SOLVENT EXTRACTION

Recommendations of solvent extraction terminology have been published by the International Union of Pure and Applied Chemistry (IUPAC) [1,3–5]. Some excellent monographs on solvent extraction are listed in section 1.8.1; unfortunately, the older ones may only be available in large science libraries. Most of them deal either with a small sector of the field or are very comprehensive. Finally, in section 1.8.2, we have collected edited versions of proceedings of the International Conferences of Solvent Extraction (ISEC), which normally are held every three years.

Because solvent extraction is used extensively as a technique in research, publications with this technique appear in all types of chemical journals. This is also true for its industrial applications that are often described in chemical engineering journals.

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Principles of Solubility and Solutions

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2.1 INTRODUCTION

Solvent extraction is another name for *liquid-liquid distribution*, that is, the distribution of a solute between two liquids that must not be completely mutually miscible. Therefore, the liquid state of aggregation of matter and the essential forces that keep certain types of liquids from being completely miscible are proper introductory subjects in a study of solvent extraction. Furthermore, the distribution of a solute depends on its preference for one or the other liquid, which is closely related to its solubility in each one of them. Thus, the general subject of solubilities is highly relevant to solvent extraction.

In a solution, the solute particles (molecules, ions) interact with solvent molecules and also, provided the concentration of the solute is sufficiently high, with other solute particles. These interactions play the major role in the distribution of a solute between the two liquid layers in liquid-liquid distribution systems. Consequently, the understanding of the physical chemistry of liquids and solutions is important to master the rich and varied field of solvent extraction.

Solvent extraction commonly takes place with an aqueous solution as one liquid and an organic solvent as the other. Obviously, the extraction process is limited to the liquid range of these substances. Since solvent extraction is generally carried out at ambient pressures, the liquid range extends from about the freezing temperature up to about the normal boiling temperature. If, however, high pressures are applied (as they are in some solvent extraction processes), then the liquid range can extend up to the critical temperature of the substance. Supercritical fluid extraction beyond the critical temperature (such as decaffeination of coffee with supercritical carbon dioxide) is a growing field of application of solvent extraction. It has the advantages that the properties of the super-critical fluid can be fine-tuned by variation of the pressure, and that this “supercritical solvent” can be readily removed by a drastic diminution of the pressure, but has drawbacks related to the high temperatures and pressures often needed.

Numerous solvents are used in solvent extraction. They can be divided in the context of solvent extraction into different classes as follows (but see Refs. [1] and [2] for some other classification schemes):

Class 1: Liquids capable of forming three-dimensional networks of strong hydrogen bonds, e.g., water, poly- and amino-alcohols, hydroxy-acids, etc.

*Retired.

Class 2: Other liquids that have both active hydrogen atoms and donor atoms (O, N, F; see Chapter. 3), but do not form three-dimensional networks (rather forming chainlike oligomers), e.g., primary alcohols, carboxylic acids, primary and secondary amines, nitro compounds with α -positioned hydrogen atoms, liquified ammonia, etc. They are generally called protic or protogenic substances.

Class 3: Liquids composed of molecules containing donor atoms, but no active hydrogen atoms, e.g., ethers, ketones, aldehydes, esters, tertiary amines, nitro compounds without α -hydrogen, phosphoryl-group containing solvents, etc. (see Table 4.3). They are generally called dipolar aprotic substances.

Class 4: Liquids composed of molecules containing active hydrogen atoms but no donor atoms, e.g., chloroform and some other aliphatic halides.

Class 5: Liquids with no hydrogen-bond forming capability and no donor atoms, e.g., hydrocarbons, carbon disulfide, carbon tetrachloride, supercritical carbon dioxide, etc.

This diversity in solvent properties results in large differences in the distribution ratios of extracted solutes. Some solvents, particularly those of class 3, readily react directly (due to their strong donor properties) with inorganic compounds and extract them without need for any additional extractant, while others (classes 4 and 5) do not dissolve salts without the aid of other extractants. These last are generally used as *diluents* for extractants, required for improving their physical properties, such as density, viscosity, etc., or to bring solid extractants into solution in a liquid phase. The class 1 type of solvents are very soluble in water and are useless for extraction of metal species, although they may find use in separations in biochemical systems (see Chapter 9).

2.1.1 Properties of Liquids

If the externally imposed conditions of pressure and temperature permit a substance to be in the liquid state of aggregation, it possesses certain general properties. Contrary to a substance in the solid state, a liquid is fluid; that is, it flows under the influence of forces and is characterized by its fluidity, or the recipro-cal, its viscosity. For water, the viscosity is only 0.89 mPa s^{-1} (nearly 1 centipoise) at 25°C [1,2], and water is considered a highly fluid liquid.

Contrary to a fluid in the gaseous state, a liquid has a surface, and is characterized by a surface tension. For water, the surface tension is 72 mN m^{-1} at 25°C [1,2]. Again, contrary to the fluid in the gaseous state, the volume of a liquid does not change appreciably under pressure; it has a low compressibility and shares this property with matter in the solid (crystalline, glassy, or amorphous) state. For water, the compressibility is $0.452 (\text{GPa})^{-1}$ at 25°C [1,2]. These are macroscopic, or bulk, properties that single out the liquid state from other states of aggregation of matter.

There are also some general microscopic, or molecular, properties that are peculiar to the liquid state. Contrary to the crystalline solid, the particles of a liquid do not possess long-range order. Although over a short range—2 to 4 molecular diameters—there is some order in the liquid, this order dissipates at longer distances. A particle in the liquid is free to diffuse and, in time, may occupy any position in the volume of the liquid, rather than being confined at or near a lattice position, as in the crystalline solid. Contrary to particles in the gaseous state, however, the particles in a liquid are in close proximity to

each other (closely packed) and exert strong forces on their neighbors [3].

The close packing of the molecules of a substance in the liquid state results in a density much higher than in the gaseous state and approaching that in the solid state. The density, ρ is the mass per unit volume, and can be expressed as the ratio of the molar mass M to the molar volume V of a liquid. Table 2.1 lists the values of the properties M and V of representative liquids that are important in the field of solution chemistry and solvent extraction. The densities and molar volumes depend on the temperature, and the latter are given for 25°C. (For a discussion of industrial solvents, see Chapter 12.)

Many liquids used in solvent extraction are polar. Their polarity is manifested by a permanent electric dipole in their molecules, since their atoms have differing electronegativities. Oxygen and nitrogen atoms, for instance, generally confer such dipolarity on a molecule, acting as the negative pole relative to carbon or hydrogen atoms bonded to them. The dipole moment μ characterizes such polar molecules and ranges from about 1.15 D (Debye unit= $3.336 \times 10^{-30} \text{ C} \cdot \text{m}$) for diethyl ether or chloroform, to 4.03 D for nitrobenzene, and 5.54 D for hexamethyl phosphoric triamide [1,2]. Water is also a polar liquid, with a moderate dipole moment, $\mu=1.83$ D. A list of dipole moments of some solvents that are important in solvent extraction is presented in Table 2.1. Substances that do not have a permanent dipole moment (i.e., $\mu=0$) are called nonpolar. Many hydrocarbons belong to this category, but not all (e.g., toluene has $\mu=0.31$ D).

When nonpolar liquids are placed in an electric field, only the electrons in their atoms respond to the external electric forces, resulting in some atomic

Table 2.1 Selected Properties of Some^a Water-Immiscible Solvents

Solvent	M g/mol	V^b mL/mol	μ^c D	ϵ^b	δ (J/mL) ^{1/2}
<i>c</i> -Hexane	84.2	108.7	0.	2.02	16.8
<i>n</i> -Hexane	86.2	131.6	0.09	1.88	15.0
<i>n</i> -Octane	114.3	163.5	~0	1.95	15.5
<i>n</i> -Decane	142.3	195.9	~0	1.99	15.8
<i>n</i> -Dodecane	170.4	228.6	~0	2.00	16.0
Decalin (mixed isomers)	138.3	157.4	0.	2.15	18.0
Benzene	78.1	89.9	0.	2.27	18.8
Toluene	92.1	106.9	0.31	2.38	18.8
Ethylbenzene	106.2	123.1	0.37	2.40	18.0
<i>p</i> -Xylene	106.2	123.9	0.	2.27	18.1
Dichloromethane	89.9	64.5	1.14	8.93	20.2
Chloroform	119.4	80.7	1.15	4.89	19.5
Carbon tetrachloride	153.8	97.1	0.	2.24	17.6
1,1-Dichloroethane	99.0	84.7	1.82	10.00	18.3
1,2-Dichloroethane	99.0	79.4	1.83	10.36	20.0

Trichloroethylene	131.4	90.7	0.80	3.42	19.0
Chlorobenzene	112.6	102.2	1.69	5.62	19.8
1,2-Dichlorobenzene	147.0	113.1	2.50	9.93	20.5
Carbon disulfide	76.1	60.6	0.	2.64	20.3
Water^a	18.0	18.1	1.85	78.36	47.9^d
Methanol ^a	32.0	40.7	2.87	32.66	29.3
Ethanol ^a	46.1	58.7	1.66	24.55	26.0
1-Propanol ^a	60.1	75.1	3.09	20.45	24.4
2-Propanol ^a	60.1	76.9	1.66	19.92	23.7
1-Butanol	74.1	92.0	1.75	17.51	23.3
Isoamyl alcohol	88.2	109.2	1.82	15.19	22.1
1-Hexanol	102.2	125.2	1.55	13.30	21.8
1-Octanol	130.2	158.4	1.76	10.34	20.9
2-Ethyl-1-hexanol	130.2	157.1	1.74	4.4	19.4
Diethyl ether	74.1	104.7	1.15	4.20	15.4
Diisopropyl ether	102.2	142.3	1.22	3.88	14.6
Bis(2-chloroethyl) ether	143.0	117.9	2.58	21.20	18.8
Acetone ^a	58.1	74.0	2.69	20.56	22.1
Methyl ethyl ketone	72.1	90.2	2.76	18.11	18.7
Methyl isobutyl ketone	100.2	125.8	2.70	13.11	17.2
Cyclohexanone	98.2	104.2	3.08	15.5	19.7
Acetylacetone	100.1	103.0	2.78	25.7	19.5
Ethyl acetate	88.1	98.5	1.78	6.02	18.2
Butyl acetate	116.2	132.5	1.84	5.01	17.6
Propylene carbonate	102.1	85.2	4.94	64.92	21.8
Nitromethane	61.0	54.0	3.56	35.87	25.7
Nitrobenzene	123.1	102.7	4.22	34.78	22.1
Acetonitrile ^a	41.1	52.9	3.92	35.94	24.1
Benzonitrile	103.1	103.1	4.18	25.2	22.7
Quinoline	129.2	118.5	2.18	8.95	22.8
Tributyl phosphate	266.3	273.8	3.07	8.91	15.3

Source: Refs. 1 and 2.

^aWater miscible.

^bAt 25°C.

^cIsolated solvent molecules, i.e., in the gaseous phase or dilute solution in an inert solvent.

^dBehaves in organic-rich aqueous mixtures as if $\delta \approx 30$.

polarization. This produces a relative permittivity (dielectric constant) ϵ , which is approximately equal to the square of the refractive index (at the sodium D line, n_D), or about 2. Polar molecules, however, further respond to the external electric field by reorienting themselves, which results in a considerably larger relative permittivity. In particular, when cooperative hydrogen bonding takes place (see later discussion), high values of ϵ are achieved (e.g., 78 for water at 25°C [1,2]). Table 2.1 lists ϵ values of some liquids that are important in solvent extraction. The attractive forces between charges (such as those on ions) of opposite sign are inversely proportional to the relative permittivity of the liquid medium in which they find themselves. Therefore, the ionic dissociation of electrolytes strongly depends on the relative permittivity of the solvent that is used to dissolve them (see section 2.6).

There are properties of liquids used as solvents, such as the liquid range, viscosity, surface tension, and vapor pressure that, although important from the practical and technical points of view, are not listed in Table 2.1. These can be found in Refs. [1,2]. Tables 12.2, 13.1, and 13.2 list solvents used in extraction with some further information.

2.1.2 Cohesive Forces in Liquids

The ability of even the most inert gases to be liquified at sufficiently low temperatures and high pressures is evidence for the existence of cohesive forces between molecules. Such forces are manifested, in their simplest form, between the atoms of noble gases, such as argon. The motions of the electrons in the argon atom induce a temporary electric dipole in a neighboring argon atom, which in turn strengthens the temporary dipole in the first one. The mutual interaction of these temporary, mutually induced electric dipoles produces attractive forces, called London forces or *dispersion forces*. These forces are rather weak, and high pressures and drastic cooling (for argon 4.9 MPa and -122°C) are required for these forces to be able to overcome the thermal agitation of the atoms that tends to keep them apart. The energy associated with the dispersion forces decreases with the sixth power of the distance between the interacting particles, and thus, these forces have a very short range, although they are present in all liquids. Hydrocarbons are often used as diluents in the organic phase of a solvent extraction system, and generally are held together as liquids by these dispersion forces operating between neighboring segments (e.g., $-\text{CH}_2-$) of different molecules. In fact, in nonpolar substances, the only cohesive forces are the dispersion forces [3].

In polar liquids, when the electric dipoles are able to arrange themselves in a “head-to-tail” configuration, that is, when their positive end is on the average more in the vicinity of the negative ends of neighboring molecules (Fig. 2.1a,c), further attractive forces result. The energy associated with these forces

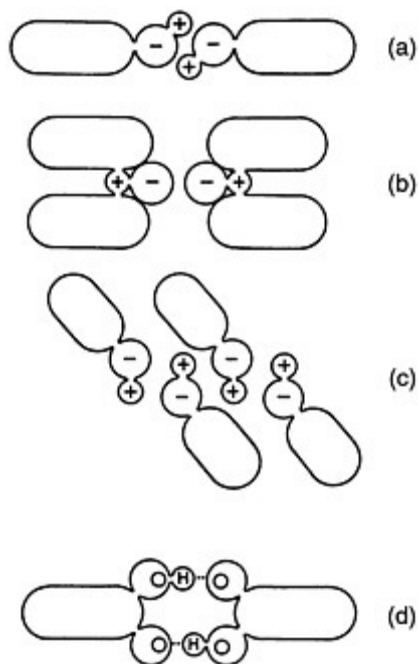


Fig. 2.1 Dipole and hydrogen bond interactions. A schematic representation of (a) “head-to-tail” dipole-dipole attractive interactions (e.g., in tri-*n*-octylamine); (b) “head-to-head” dipole-dipole repulsive interactions caused by steric hindrance (e.g., in dibutyl sulfoxide); (c) chainlike dipole-dipole interactions (e.g., in 1-octanol); (d) a cyclic, hydrogen-bonded dimer (e.g., in hexanoic acid).

is proportional to the product of the dipole moments of the molecules and is inversely proportional to the sixth power of the mean distance between them (i.e., it has the same range as the dispersion forces). However, the structure of the molecules may be such that they prevent a head-to-tail configuration (see Fig. 2.1b), and the resulting head-to-head configuration causes repulsion between the polar molecules [3].

Some of the liquids that are used in solvent extraction, especially water, interact by means of *hydrogen bonding*. Their molecules have a hydrogen atom attached to a very electronegative element (mainly oxygen and, less effectively, nitrogen), and this hydrogen atom is able to be bound to the electronegative atom (O, N, or F) of a neighboring molecule, forming a hydrogen bridge. Such a bridge requires a rather rigid geometry: a linear configuration of the two electronegative atoms of the neighboring molecules with the hydrogen atom between them. This bond is of considerably greater strength than dispersion and dipole-dipole interactions. If the molecules of a substance can both donate and accept a hydrogen bond, a cyclic dimer may result (see Fig. 2.1d) that is considerably less polar than the monomers of this substance. Occasionally, water again being an example, this hydrogen bonding is a cooperative phenomenon: if such a

bond is formed between two molecules, each becomes more likely to participate in further hydrogen bonding with other neighbors [3].

The three kinds of forces described above, collectively known as the cohesive forces that keep the molecules of liquids together, are responsible for various properties of the liquids. In particular, they are responsible for the work that has to be invested to remove molecules from the liquid, that is, to vaporize it. The energy of vaporization of a mole of liquid equals its molar heat of vaporization, $\Delta_v H$, minus the pressure-volume work involved, which can be approximated well by $\mathbf{R}T$, where $\mathbf{R}$ is the gas constant [$8.3143 \text{ J K}^{-1} \text{ mol}^{-1}$] and T is the absolute temperature. The ratio of this quantity to the molar volume of the liquid is its *cohesive energy density*. The square root of the cohesive energy density is called the (Hildebrand) solubility parameter of the liquid, δ :

$$\delta^2 = (\Delta_v H - \mathbf{R}T) / V \quad (2.1)$$

where V is the molar volume. A list of solubility parameters of representative liquids that are important in solvent extraction is given in Table 2.1.

2.2 SOLVENT MISCIBILITY

Solvent extraction takes place through the distribution of a solute or of solutes between two practically immiscible liquids. For a separation to be carried out by solvent extraction, the solute has to transfer from one region of space to another such region, which is physically separated from the first (see Fig. 1.1). In each such region, the solute is dissolved uniformly in a (*homogeneous*) liquid and, at all points, is under the same pressure. Such a region of space is called a *phase*. At equilibrium, at a constant temperature and pressure, a liquid phase is homogeneous and is isotropic: it has the same properties in all directions. In solvent extraction, we are interested in two liquid phases in contact, normally under the same temperature and pressure, which eventually reach *mass transfer equilibrium*. Such an equilibrium is a dynamic process: as many molecules of one solvent enter the other as leave it and return to their original phase as do the molecules (ions) of the solute(s). If each of the two liquids is originally a pure chemical substance and there is (as yet) no solute to distribute, we speak of a *two-component*, two-phase system, but the net effect is that an amount of the substance from one liquid phase is transferred to the other and vice versa. For two practically immiscible liquids this transferred amount is small relative to the total amount of the two phases. There may or may not be a vapor phase present, but this is generally disregarded in the context of solvent extraction.

2.2.1 The Phase Rule

A substance that can be added to a system independently (or removed from it, say by precipitation or vaporization) is called a *component* of such a system. The *phase rule*, summarizing a general behavior of nature, says:

The number of degrees of freedom in a system equals the number of components plus two minus the number of phases.

A two-component, two-phase system, therefore, has two degrees of freedom, that is, two external conditions, such as the temperature and pressure, specify its composition completely. By this we mean that the composition at equilibrium of each of the phases is uniquely determined at a given temperature and pressure. For example, in the diethylether-water system at atmospheric pressure and at 10°C, the ether-rich layer contains 1.164 mass% water (4.6 mol% water) and the water-rich layer contains 9.04 mass% ether (2.4 mol% ether). This holds irrespective of the relative amounts of the liquid phases present, as long as there are two phases. When the temperature is raised to 30°C, the composition of the two phases changes to 1.409 mass% water (5.5 mol% water) and 5.34 mass% ether (1.35 mol% ether) in the two phases, respectively. These compositions are uniquely determined by the (atmospheric) pressure and the temperature, two external conditions (degrees of freedom) for this two-component, two-phase system. [If a third phase, the vapor, is present, then only one degree of freedom remains, and, e.g., the temperature determines the (vapor) pressure attained by the system]. For a two-phase system, each additional component (e.g., a solute) permits a new degree of freedom: here, the concentration of the solute in one of the phases. When this has been fixed, its concentration in the other phase is given by the equilibrium condition and cannot be freely chosen.

For the two-component, two-phase liquid system, the question arises as to how much of each of the pure liquid components dissolves in the other at equilibrium. Indeed, some pairs of liquids are so soluble in each other that they become completely miscible with each other when mixed at any proportions. Such pairs, for example, are water and 1-propanol or benzene and carbon tetrachloride. Other pairs of liquids are practically insoluble in each other, as, for example, water and carbon tetrachloride. Finally, there are pairs of liquids that are completely miscible at certain temperatures, but not at others. For example, water and triethylamine are miscible below 18°C, but not above. Such pairs of liquids are said to have a *critical solution temperature*, T_{cs} . For some pairs of liquids, there is a lower T_{cs} (LCST), as in the water-triethylamine pair, but the more common behavior is for pairs of liquids to have an upper T_{cs} (UCST), (Fig. 2.2) and some may even have a closed mutual solubility loop [3]. Such instances are rare in solvent extraction practice, but have been exploited in some systems, where separations have been affected by changes in the temperature.

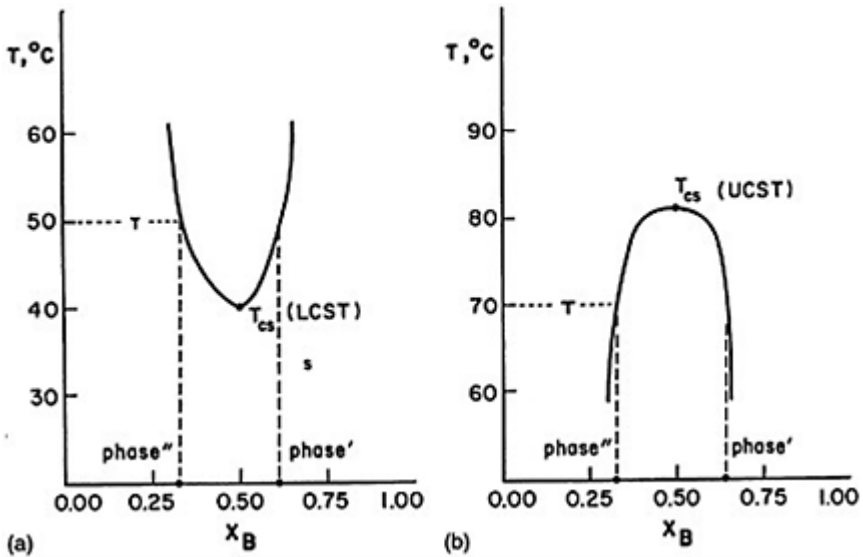


Fig. 2.2 Liquid immiscibility. The guaiacol (A)+glycerol (B) system happens to have a closed miscibility loop. The (phase) coexistence curves are shown on the left-hand side (a) for lower temperatures, at which a lower critical solution temperature (LCST), $T_{cs} = 40^\circ\text{C}$, is seen, and on the right-hand side (b) for higher temperatures, where a UCST, $T_{cs} = 82^\circ\text{C}$, is seen. The compositions of the A-rich phases'' and the B-rich phases' are shown at 50°C and 70°C , respectively.

2.2.2 Excess Gibbs Energy of Mixing

The mutual solubility of two liquids A and B depends, in general, on how much the molecules of each liquid tend to attract those of its own kind, relative to their tendency to attract those of the other. This tendency is measured by the excess Gibbs energy of mixing of the two liquids (see section 2.4), $\Delta_M G_{AB}^E$, which is related to the partial vapor pressures p_A and p_B of the two liquids A and B in the mixture. If the composition of the system is given by n_A and n_B moles of the respective components in a given phase, their *mole fractions* in this phase are

$$x_A = n_A / (n_A + n_B) \quad x_B = 1 - x_A = n_B / (n_A + n_B) \quad (2.2)$$

The excess Gibbs energy of mixing per mole of mixture is approximately

$$\Delta_M G_{AB}^E = RT [x_A \ln (p_A / x_A p_A^*) + x_B \ln (p_B / x_B p_B^*)] \quad (2.3)$$

where p_A^* and p_B^* are the vapor pressures of the pure liquids A and B, respectively, at the given temperature T . (The approximation consists of neglecting the interactions in the vapors that is permissible at low vapor pressures.)

The system of two liquids will split into two liquid phases at a given T and composition x if

$$x_A x_B (\partial^2 \Delta_M G_{AB}^E / \partial x^2) < -RT \quad (2.4)$$

Many liquid mixtures behave approximately as *regular mixtures* [3], which are characterized by

$$\Delta_M G_{AB}^E = x_A x_B b_{AB}(T) \quad (2.5)$$

$b_{AB}(T)$ being a (generally temperature-dependent) constant that is independent of the composition. Such regular mixtures split into two liquid phases if the following inequality is obeyed:

$$b_{AB}(T) > RT / (2x_A x_B) \quad (2.6)$$

They generally exhibit a UCST, $T_{cs} = b_{AB}(T) / 2R$. It must be remembered that only pairs of liquids that mix at least partly behave as regular mixtures and that Eqs. (2.5) and (2.6) cannot be used for the general case of any two liquids, except as a very rough approximation. The value of $b_{AB}(T)$ can be estimated from the properties of the individual liquids to be mixed, by invoking their solubility parameters [Eq. (2.1)]. The expression is:

$$b_{AB}(T) = V_A^* V_B^* (x_A V_A^* + x_B V_B^*)^{-1} (\delta_A^2 - \delta_B^2) \quad (2.7)$$

where V_A^* and V_B^* are the respective molar volumes of the neat liquids. [Eq. (2.7) makes $b_{AB}(T)$ composition dependent and should be used only if the molar volumes are not too disparate, since otherwise the mixture is not a regular one.] Thus, the larger the difference between the solubility parameters of the two liquids, the more likely is $b_{AB}(T)$ to be larger than the right-hand side of Eq. (2.6), and the larger is the tendency of the two-component system to split into two liquid phases [3].

If the mutual miscibility is small, then approximately

$$\ln x_A' \approx -V_B^* (\delta_A - \delta_B)^2 / RT \quad (2.8)$$

where x_A' is the mole fraction of component A in the B-rich phase marked by '. A similar

equation holds for x_B'' in the A-rich phase, marked " , with V_A^* replacing V_B^* . Thus, the composition of the two phases can also be estimated by the application of the solubility parameters. These considerations can also be used for an estimate of the distribution of a solute between the immiscible liquids A and B, assuming the conditions for regular mixing to be fulfilled for all the components (see section 2.8.2).

As an example consider the ethylene glycol (A)-benzene (B) system at 25 °C, assuming it to be a regular mixture (actually it is not). The relevant quantities are $V_A^*=55.9$ and $V_B^*=89.9 \text{ cm}^3 \text{ mol}^{-1}$, $\delta_A=32.4$ and $\delta_B=16.0 \text{ J}^{1/2} \text{ cm}^{-3/2}$. Hence for the equimolar mixture $x_A=x_B=0.5$, according to Eq. (2.7), $b_{AB}=54.7 \text{ kJ mol}^{-1} \gg RT/2x_Ax_B=5 \text{ kJ mol}^{-1}$, so that this system ought to split into two liquid phases, as it does. In the benzene-rich phase (') the ethylene glycol content should be 0.005 wt% according to Eq. (2.8) and the benzene content in the ethylene glycol-rich phase (") should be 0.31 wt%. Most regular mixtures, however, would form a single homogeneous liquid phase, since the condition of Eq. (2.6) is rarely met.

2.2.3 The Mutual Solubility of Solvents

An inspection of Table 2.1 shows that, of the common liquids, water has by far the largest solubility parameter, $\sim 48 \text{ J}^{1/2} \text{ cm}^{-3/2}$ [1,2]. Many organic solvents have appreciably smaller values of δ . Pairs of liquids that have values of δ that are less than 4 units apart have a reasonably large mutual solubility. They may even be completely miscible. Thus, aliphatic hydrocarbons mix with one another or with aromatic hydrocarbons to form a single homogeneous phase, as they also do with most halogenated hydrocarbons. Pairs of liquids having values of δ that are much more than 4 units apart are partially or nearly completely immiscible. This, however, does not apply well when one of the pair is water, unless it is assigned the value $\delta \cong 30 \text{ J}^{1/2} \text{ cm}^{-3/2}$, which is comparable with that of methanol.

Water mixes completely with the lower alcohols (methanol, ethanol, 1-and 2-propanol, and *t*-butanol), as well as with most polyhydric alcohols and the lower ether alcohols (2-methoxy-ethanol, and so on). Of the ethers, water is miscible with tetrahydrofuran, 1,2-dimethoxyethane, and 1,4-dioxane, but not with the higher members. Among the ketones, water is miscible with acetone only; among the carboxylic acids, water is miscible with those up to and including butanoic acid, but with none of the common esters (an exception being ethylene carbonate). Among nitrogen-containing organic liquid compounds, water is miscible with acetonitrile, but not the higher ones (except dinitriles, such as succinonitrile). Water is miscible with primary, secondary, and tertiary amines, up to a total of about five carbon atoms in all the chains, including pyridine, but not with nitro compounds. Water is also miscible with many difunctional liquid compounds, such as diamines, alcohol amines, amides, and others, and with such compounds as dimethylsulfoxide [1–3].

The higher members of homologous series based on these compounds become increasingly immiscible with water as the chain length increases or as aromatic rings are added. Consequently, these higher members can be used as solvents in extraction systems. Table 2.2 lists the mutual solubilities, in weight percent (wt%), of water and

representative organic solvents at 25°C, unless noted otherwise.

2.3 SOLUTE-SOLVENT INTERACTIONS

When a solute particle is introduced into a liquid, it interacts with the solvent particles in its environment. The totality of these interactions is called the *solvation* of the solute in the particular solvent. When the solvent happens to be water, the term used is *hydration*. The solvation process has certain consequences pertaining to the energy, the volume, the fluidity, the electrical conductivity, and the spectroscopic properties of the solute-solvent system. The apparent molar properties of the solute ascribe to the solute itself the entire change in the properties of the system that occur when 1 mol of solute is added to an infinite amount of solution of specified composition. The solvent is treated in the calculation of the apparent molar quantities of the solute as if it had the properties of the pure solvent, present at its nominal amount in the solution. The magnitudes of quantities, such as the apparent molar volume or heat content, do convey some information on the system. However, it must be realized that both the solute and the solvent are affected by the solvation process, and more useful information is gained when the changes occurring in both are taken into account.

2.3.1 Interactions at the Molecular Scale

The solvation process can be envisioned as occurring in several stages, although only the sum of the stage contributions to the overall process is measurable. First, a cavity must be created in the solvent to accommodate the solute. Then the solute is placed in the cavity and permitted to interact with its nearest neighbors, eventually forming coordinate bonds with some of them, forming a new entity, the solvated solute. Finally, this entity may interact further with its surroundings, by orienting solvent molecules, by the formation or disruption of hydrogen bonds, or by other interactions.

If the solute is charged (i.e., if it is an ion), it will orient the dipoles of a polar solvent by its electrical field. Donor-acceptor bonds may be formed if the

Table 2.2 Mutual Solubility of Water and Some Organic Solvents at 25°C

Solvent	Solvent in water wt%	Water in solvent wt%
<i>c</i> -Hexane	0.0055	0.010 ^a
<i>n</i> -Hexane	0.00123	0.0111 ^a
<i>n</i> -Octane	6.6×10^{-7}	0.0095 ^a
<i>n</i> -Decane	5.2×10^{-8}	7.2×10^{-5}
<i>n</i> -Dodecane	3.7×10^{-9}	6.5×10^{-5}
Decalin (mixed isomers)	<0.02	0.0063 ^a
Benzene	0.179	0.0635

Toluene	0.0515	0.0334
Ethylbenzene	0.0152	0.043
<i>p</i> -Xylene	0.0156	0.0456
Dichloromethane	1.30	0.198
Chloroform	0.815 ^a	0.093
Carbon tetrachloride	0.077	0.0135 ^b
1,1-Dichloroethane	5.03 ^a	0.096
1,2-Dichloroethane	0.81 ^a	0.187
Trichloroethylene	0.137	0.32
Chlorobenzene	0.0488 ^b	0.0327
1,2-Dichlorobenzene	0.0156	0.309
Carbon disulfide	0.210 ^a	0.0142
1-Butanol	7.45	20.5
Isobutyl alcohol	10	16.9
1-Pentanol	2.19	7.46
Isoamyl alcohol	2.97	9.19
1-Hexanol	0.7061	7.42
1-Octanol	0.0538	
2-Ethyl-1-hexanol	0.07 ^a	2.6 ^a
Phenol	8.66	28.72
<i>m</i> -Cresol	2.51 ^c	
Diethyl ether	6.04	1.468
Diisopropyl ether	1.2 ^a	0.57 ^a
Bis(2-chloroethyl) ether	1.02 ^a	0.1 ^a
Methyl ethyl ketone	24.0 ^a	10.0 ^a
Methyl isobutyl ketone	1.7	1.9
Cyclohexanone	2.3 ^a	8.0 ^a
Acetylacetone	16.6 ^a	4.5 ^a
Ethyl acetate	8.08	2.94
Butyl acetate	0.68 ^a	1.2 ^a
Propylene carbonate	17.5	8.3
Nitromethane	11.1	2.09
Nitrobenzene	0.19 ^a	0.24 ^a
Benzonitrile	0.2	1 ^b
Quinoline	0.609 ^a	
Tri- <i>n</i> -butyl phosphate	0.039	4.67

Note. See Table 2.1 for some miscible solvents, marked by ^a.

^aAt 20°C.

^bAt 30°C.

^cAt 40°C.

solute and solvent have suitable electron pair donation and acceptance properties.

Hydrogen bonds between the solute and neighboring solvent molecules may form if one or the other or both are *protic*, i.e., have hydrogen atoms bonded to electronegative atoms. The solute particle may also undergo changes in the solvation process, its internal structure, if it has one, being affected by the strengthening or weakening of certain bonds, by a redistribution of the partial electrical charges on its atoms, or by the favoring of a certain conformation of a flexible solute molecule.

Spectroscopic measurements may, in certain cases, yield direct information on these interactions. On the other hand, thermodynamic values, obtained by measuring certain bulk properties of the system, require the aid of statistical mechanical methods to be related to specific interactions between the solute and the solvent. However, the thermodynamic aspects of the solute-solvent interactions reflect the preference of the solute for one solvent over another and, thereby, determine distribution of the solute in a solvent extraction system.

2.3.2 Thermodynamics of the Interactions

What quantity describes best the totality of these solute-solvent interactions and how can the various contributions to them be estimated? Let the pure solute B be vaporized in an imaginary process to a gas, and let this gas be very dilute, so that it obeys the *ideal gas laws*. In this condition each particle of the solute (molecule or ion) is very remote from any neighbor and has no environment with which to interact. If B is polyatomic, it does have its internal degrees of freedom, such as bond vibrations and rotation of the particle.

If this ideal gaseous particle B is introduced into a liquid A at a given temperature and pressure, all of the solute-solvent interactions become “switched on,” the solvation process takes place, and the *Gibbs energy of solvation*, $\Delta_{\text{soln}} G_B$, is released. In many cases the process of dissolution of a gaseous solute in a liquid solvent can, indeed, be carried out experimentally—for instance, when propane or carbon dioxide is dissolved in water to give a solution at a given gas pressure.

The Gibbs energy change for the process of dissolution of the gaseous solute B, $\Delta_{\text{soln}} G_B$, is the driving force for the material transfer. When equilibrium is reached, $\Delta_{\text{soln}} G_B$ becomes zero (since, at equilibrium, no more net transfer occurs). The following equation then holds:

$$\Delta_{\text{soln}} G_{B \text{ eq}} = 0 = \Delta_{\text{soln}} G_B^\circ + RT \ln [c_B(1) / c_B(g)]_{\text{eq}} \quad (2.9)$$

where $\Delta_{\text{soln}} G_B^\circ$ is the *standard molar Gibbs energy of solvation* of the solute B, defined by Eq. (2.9) and c_B is its molar concentration (moles per unit volume) in the designated phases, (1) for liquid and (g) for gaseous. Equilibrium is generally assumed for the processes discussed elsewhere in this chapter, and is only emphasized here for Eq. (2.9) by the subscript eq for $\Delta_{\text{soln}} G_B$ and the ratio of concentrations, $c_B(1)/c_B(g)$. The concentration in the gas phase is, of course, given by the pressure according to the ideal gas law $c_B(g) = n_B/V = P/RT$. The ratio of the concentrations is known as the *Bunsen coefficient*, $K_{B(B,A)}$, for the solubility of the gaseous solute B in liquid A. At low pressures and concentrations $K_{B(B,A)}$ depends only on the temperature and is independent

of the pressure and the concentration.

Whether obtained from an actual experimentally feasible process or from a thought process, $\Delta_{\text{soln}} G_B^\circ$ which is obtained from Eq. (2.9) by re-arrangement, pertains to the solvation of the solute and expresses the totality of the solute-solvent interactions. It is a *thermodynamic function of state*, and so are its derivatives with respect to the temperature (the standard molar entropy of solvation) or pressure. This means that it is immaterial how the process is carried out, and only the initial state (the ideal gaseous solute B and the pure liquid solvent) and the final state (the dilute solution of B in the liquid) must be specified.

Because it is a function of state, $\Delta_{\text{soln}} G_B^\circ$ may be considered to be made up additively of the contributions from the various stages in which the transfer of the solute particle from the gaseous state into the liquid solvent has been envisaged by the foregoing to take place.

The first stage is the creation of the cavity in the liquid to accommodate the solute. Obviously, work must be done against the cohesive forces of the liquid that hold its molecules together. This work should be proportional to the required size of the cavity, and increase as the volume of the solute increases. An expression for this work would be

$$\Delta_{\text{cav}} G = A_{\text{cav}} V_B \delta_A^2 \quad (2.10)$$

where A_{cav} is a proportionality coefficient, V_B is the molar volume of the solute B, and δ_A^2 is the cohesive energy density of the solvent A. Thus, in a series of solvents for a given solute, the positive contribution of cavity formation to $\Delta_{\text{soln}} G_B^\circ$ increases with the squares of the solubility parameters of the solvents. For a series of solutes and a given solvent, it increases with the molar volumes of the solutes.

It is more difficult to estimate the contribution from the dispersion forces to the solute-solvent interactions. Their energy increases with the product of the polarizabilities of the partners, but decreases strongly with the distance between them (being proportional to the inverse sixth power of the distance). The polarizability is related to the molar volume, hence, to the third power of the linear dimension of the solute or the solvent. Hence, the product of the polarizabilities depends on the sixth power of the distance between the centers of the interacting molecules. Consequently, these tendencies balance each other. For large molecules (e.g., metal chelates, liquid hydrocarbons), it is better to consider the interactions between adjacent segments of neighboring molecules, which are at a constant mean distance from one another in the liquid solution. The contribution from the dispersion forces (negative, because they are attractive) is proportional to the surface areas of the interacting molecules, or to the number N of segments present, and depends on their chemical natures. If A represents again the liquid solvent, then [3]

$$\Delta_{\text{disp}} G = \sum A_{\text{iBA}} N_{\text{iB}} N_{\text{iA}} \quad (2.11)$$

where A_{iBA} is the (negative) interaction Gibbs energy of a pair of segments of kind i , and the summation extends over all the different kinds of segments. A methylene group, a halogen atom, a —CH=CH— group of an aromatic ring, or some other functional group, generally serves as a segment.

There may be additional, specific interactions between the solute and the solvent. Hydrogen bonds may be formed between them, particularly in protic solvents, i.e., solvents that contain hydrogen atoms bonded to oxygen (more rarely, nitrogen) atoms, such as water, alcohols, carboxylic acids, or acidic phosphoric esters. Hydrogen bonds are formed with solute anions, with hydrated ions in general (having an outer surface of water molecules), and with neutral solutes that have a very basic atom with a lone pair of electrons that can accept a hydrogen bond. Also, donor-acceptor bonds can be formed if the solvent has a very basic atom in its structure donating a pair of unshared electrons, if it is suitably exposed, and the solute is a cation or some acidic neutral molecule accepting this pair.

A generalized equation for $\Delta_{\text{solv}} G_B^\circ$ is [4]

$$\Delta_{\text{solv}} G_B^\circ = A_0 + A_\pi \pi^* + A_\alpha \alpha + A_\beta \beta + A_\delta \delta^2 \quad (2.12)$$

which describes the value of $\Delta_{\text{solv}} G_B^\circ$ for a given solute (characterized by A_0 , A_π , A_α , A_β , and A_δ) with a series of solvents. The solvents are characterized by their Taft-Kamlet *solvatochromic parameters*: π^* for polarity-polarizability, α for hydrogen bond donation acidity, and β for hydrogen bond acceptance basicity. Values of these solvatochromic parameters have been tabulated for many solvents. (Table 2.3 gives the values for selected solvents.) As before, δ is the solubility parameter. The first two terms on the right-hand side of Eq. (2.12) express $\Delta_{\text{disp}} G$, the next two the hydrogen-bonding interactions, and the last one $\Delta_{\text{cav}} G$.

As an example, consider phenol as the solute and water and toluene as two solvents. The parameters for phenol are $A_\pi=5.7$, $A_\alpha=-12.9$, $A_\beta=-18.3$, and $A_\delta=0.0091$, whereas A_0 is unspecified, but a negative quantity. With the solvent parameters from Tables 2.1 and 2.3, the standard Gibbs energy of solvation of phenol in water becomes $A_0+3.39$, and in toluene $A_0+4.11 \text{ kJ mol}^{-1}$. It is seen that $\Delta_{\text{solv}} G_B^\circ$ is lower in water than in toluene, so that the transfer of phenol from water to toluene entails an increase in $\Delta_{\text{solv}} G_B^\circ$. The consequence of this is that phenol prefers water over toluene, since work would be required to make this transfer. It should be remembered that the standard Gibbs energies of solvation refer to the state of infinite dilution of the solute (solute-solute

Table 2.3 Solvatochromic Parameters for Some Solvents and (Monomeric) Solutes (in Parentheses)

Substance	π^*	α	β	Substance	π^*	α	β
<i>c</i> -Hexane	0	0	0	2-Butanone	0.60	0.06	0.48
<i>n</i> -Hexane	-0.11	0	0	Cyclohexanone	0.68	0	0.53
Benzene	0.55	0	0.10	Acetophenone	0.81	0.04	0.49
Toluene	0.49	0	0.11	Acetic acid	0.64	1.12	0.45
<i>p</i> -Xylene	0.45	0	0.12	Hexanoic acid	0.52	1.22	0.45
Dichloromethane	0.82	0.13	0.10	Benzoic acid	0.80	(0.87)	(0.40)
Chloroform	0.58	0.20	0.10	Ethyl acetate	0.45	0	0.45
Carbon tetrachloride	0.21	0	0.10	Butyl acetate	0.46	0	0.45
1,2-Dichloroethane	0.73	0	0.10	Propylene carbonate	0.83	0	0.40
Chlorobenzene	0.68	0	0.07	Formamide	0.97	0.71	0.48
Water	1.09	1.17	0.47	Dimethylformamide	0.88	0	0.69
Water monomer	(0.39)	(0.32)	(0.15)	Tetramethylurea	0.79	0	0.80
Methanol	0.60	0.98	0.66	Hexamethyl phosphoramide	0.87	0	1.00
Methanol monomer	(0.39)	(0.38)	(0.41)	Acetonitrile	0.66	0.19	0.40
Ethanol	0.54	0.86	0.75	Benzonitrile	0.88	0	0.37
Ethanol monomer	(0.39)	(0.36)	(0.47)	Nitromethane	0.75	0.22	0.06
1-Propanol	0.52	0.84	0.90	Nitrobenzene	0.86	0	0.30
2-Propanol	0.48	0.76	0.84	Triethylamine	0.09	00.71	
1-Hexanol	0.40	0.80	0.84	Tri- <i>n</i> -butylamine	0.06	0	0.62
Trifluoroethanol	0.73	1.51	0	Dimethyl sulfoxide	1.00	0	0.76
Trifluoroethanol monomer		(0.59)	(0.10)	Diphenyl sulfoxide		0	0.70
Phenol	0.72	1.65	0.30	Sulfolane	0.90	0	0.39
Diethyl ether	0.24	0	0.47	Triethyl phosphate	0.69	0	0.77
Diisopropyl ether	0.19	0	0.49	Tributyl phosphate	0.63	0	0.80
Tetrahydrofuran	0.55	0	0.55	Triethyl phosphine oxide		0	1.05
Dioxane	0.49	0	0.37	Triphenyl phosphine oxide		0	0.94
Anisole	0.70	0	0.32	Pyridine	0.87	0	0.64
1,2-Dimethoxyethane	0.53	0	0.41	Quinoline	0.93	0	0.64
Acetone	0.62	0.08	0.48				

Source: Ref. 2.

interactions being, therefore, absent), and that a reaction such as ionic dissociation of the phenol in water is ignored in this example.

Extensive tables of solute parameters are beyond the scope of this book. Equations (2.10) to (2.12) are meant to show the nature of the dependencies of the additive terms on various quantities. They enable the prediction of tendencies of solute-solvent interactions for a given solute with a series of solvents or for a series of solutes with a given solvent.

2.3.3 Solvation of Electrolytes

If the solute B is an electrolyte that dissociates into ions in the solvent in question, the solvation of each ion separately may be considered in the hypothetical transfer process, although experimentally only quantities pertaining to the entire electrolyte can be measured [5]. The ions of a crystalline salt are brought from their standard solid state into the ideal gas state by the investment of the lattice Gibbs energy (approximately the lattice energy for a salt dissociating into one cation and one anion). The transfer of an ion from the ideal gas to the liquid is accompanied by reorientation of the dipoles of the solvent molecules. The contribution to $\Delta_{\text{solv}}G_{\text{B}}^{\circ}$ of this reorientation is given by the *Born equation*:

$$\Delta_{\text{el}}G = -A_{\text{el}}z^2r^{-1}(1-1/\epsilon) \quad (2.13)$$

where $A_{\text{el}} = N_{\text{A}}e^2/8\pi\epsilon_0 = 69.5 \text{ kJ mol}^{-1}$ (with r in nm), z is the charge on the ion in proton charge units, r is a distance related to the radius of the ion in nm (Table 2.4), and ϵ is the relative permittivity (dielectric constant) of the solvent. The Born equation should be applied only from a distance r from the center of the ion that is larger than the radius of the bare ion by Δr , which depends on the sizes of the ion B and the solvent molecules, A. Closer to the ion, the phenomenon of dielectric saturation sets in (Fig. 2.3), and the solvent cannot reorient itself freely because the binding to the ion becomes too strong. The contribution from this *inner shell of solvation* to $\Delta_{\text{solv}}G_{\text{B}}^{\circ}$ must be estimated by other means (see, e.g., Ref. [5]), but for polar solvents can be taken to be only slightly dependent on the nature of the solvent (except that it may depend on the thickness of this inner shell). In any event, Eq. (2.13) shows that the contribution of the reorientation of the solvent by the ion depends on the square of the charge on the ion and, reciprocally, on both the size of the ion and the relative permittivity of the solvent.

Terms for the electrostatic interactions [Eq. (2.13)] for the region outside the first solvation shell, and an appropriate one for the inner region, must be added to Eq. (2.12) for each ion of an electrolyte B, for the evaluation of $\Delta_{\text{solv}}G_{\text{B}}^{\circ}$. Since cations do not accept hydrogen bonds and anions do not donate them, except when protonated, like HSO_4^- , the term in α of the solvent becomes unimportant for cations and that in β of the solvent for anions.

Table 2.4 Crystal Ionic Radii and Standard Molar Gibbs Free Energies of Hydration of Ions

Ion	r (nm)	$-\Delta_{\text{hyd}}G^\circ$ (kJ/mol)	Ion	r (nm)	$-\Delta_{\text{hyd}}G^\circ$ (kJ/mol)	Ion	r (nm)	$-\Delta_{\text{hyd}}G^\circ$ (kJ/mol)
H ⁺		1056				F ⁻	0.133	472
Li ⁺	0.069	481	Al ³⁺	0.053	4531	Cl ⁻	0.181	347
Na ⁺	0.102	375	Sc ³⁺	0.075	3801	Br ⁻	0.196	321
K ⁺	0.138	304	Y ³⁺	0.090	3457	I ⁻	0.220	283
Rb ⁺	0.149	281	La ³⁺	0.105	3155	OH ⁻	0.133	439
Cs ⁺	0.170	258	Ce ³⁺	0.101	3209	SH ⁻	0.207	303
NH ₄ ⁺	0.148	292	Nd ³⁺	0.099	3287	CN ⁻	0.191	305
Me ₄ N ⁺	0.280	412	Gd ³⁺	0.094	3385	SCN ⁻	0.213	287
Ag ⁺	0.115	440	Ho ³⁺	0.090	3480	N ₃ ⁻	0.195	287
Ph ₄ As ⁺	0.425	-32	Lu ³⁺	0.086	3522	BF ₄ ⁻	0.230	200
Mg ²⁺	0.072	1838	Pu ³⁺	0.101	3245	ClO ₃ ⁻	0.200	287
Ca ²⁺	0.100	1515	Am ³⁺	0.100	3297	BrO ₃ ⁻	0.191	340
Sr ²⁺	0.113	1386	Cr ³⁺	0.062	4010	IO ₃ ⁻	0.181	408
Ba ²⁺	0.136	1258	Fe ³⁺	0.065	4271	ClO ₄ ⁻	0.240	214
Mn ²⁺	0.083	1770	Ga ³⁺	0.062	4521	MnO ₄ ⁻	0.240	245
Fe ²⁺	0.078	1848	In ³⁺	0.079	3989	NO ₂ ⁻	0.192	339
Co ²⁺	0.075	1922	Tl ³⁺	0.088	3976	NO ₃ ⁻	0.179	306
Ni ²⁺	0.069	1992	Bi ³⁺	0.102	3486	CH ₃ CO ₂ ⁻	0.232	373
Cu ²⁺	0.073	2016	Ce ⁴⁺	0.080	6129	BPh ₄ ⁻	0.421	-42
Zn ²⁺	0.075	1963	Th ⁴⁺	0.100	5823	C ₂ O ₄ ²⁻	0.178	1479
Cd ²⁺	0.095	1763	U ⁴⁺	0.097	6368	C ₂ O ₄ ²⁻	0.21	1200
Hg ²⁺	0.102	1766	Zr ⁴⁺	0.072	6799	SO ₃ ²⁻	0.20	1303
Pb ²⁺	0.118	1434	Hf ⁴⁺	0.071	6975	SO ₄ ²⁻	0.230	1090
UO ₂ ²⁺	0.28	1229				CrO ₄ ²⁻	0.240	958
						PO ₄ ³⁻	0.238	2773

Source: Ref. 6.

These considerations are important, since solvent extraction often involves aqueous ions that must be transferred from an aqueous solution to some organic solvent. (The latter may contain complexing reagents that interact with the ions.) This transfer can be envisaged as taking place by steps in which the first one involves the freeing of an ion from its hydration before it is permitted to react with the solvent or reagent into which it is transferred. This freeing requires the investment of work that is the negative of the standard Gibbs energy of hydration of the ion [5,6]. The specification of this quantity requires splitting the value of an appropriate electrolyte into individual values for the cation and the anion. This cannot be accomplished by purely thermodynamic means, and

an extrathermodynamic assumption or convention is needed. Several such assump-

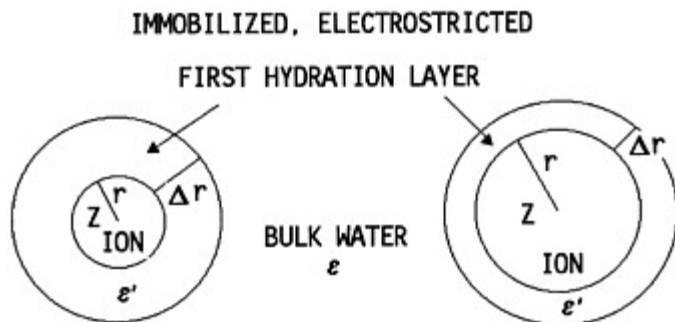


Fig. 2.3 A schematic representation of the hydration layer near a small ion (left) and a large ion (right), showing the region where the water is dielectrically saturated (with a low relative permittivity ϵ'), hence electrostricted (squeezed) and immobilized. The thickness of this layer, Δr , depends reciprocally on the size of the ion.

tions that lead to consistent results have been proposed. One of them involves a *reference electrolyte*, of which the cation and anion are large, spherical, univalent, and similar in all respects except the sign of the charge. Such a reference electrolyte is tetraphenylarsonium tetraphenylborate (TATB); the standard molar Gibbs energy of hydration of each of its ions is $\Delta_{\text{hyd}}G^\circ = -38 \pm 6 \text{ kJ mol}^{-1}$. On this basis, values of $\Delta_{\text{hyd}}G^\circ$ of other ions can be obtained, consistent with certain other extrathermodynamic assumptions, and are listed in Table 2.4 [6].

The differences in the solvation abilities of ions by various solvents are seen, in principle, when the corresponding values of $\Delta_{\text{solv}}G^\circ$ of the ions are compared. However, such differences are brought out better by a consideration of the standard molar Gibbs energies of transfer, $\Delta_t G^\circ$ of the ions from a reference solvent into the solvents in question (see further section 2.6.1). In view of the extensive information shown in Table 2.4, it is natural that water is selected as the *reference solvent*. The TATB reference electrolyte is again employed to split experimental values of $\Delta_t G^\circ$ of electrolytes into the values for individual ions. Tables of such values have been published [5–7], but are outside the scope of this text. The notion of the standard molar Gibbs energy of transfer is not limited to electrolytes or ions and can be applied to other kinds of solutes as well. This is further discussed in connection with solubilities in section 2.7.

2.4 THERMODYNAMICS OF SOLUTIONS

Thermodynamics is the branch of science dealing with the energetics of substances and processes. It describes the tendency of processes to take place spontaneously, the effects of external conditions, and the effects of the composition of mixtures on such processes. Thermodynamics is generally capable of correlating a variety of data pertaining to

widely changing conditions by relatively simple formulae. One approach to such a correlation involves the definition of a hypothetical ideal system and the subsequent consideration of deviations of real systems from the ideal one. In many cases, indeed, such deviations are relatively small and can be ignored in a first approximation. Such examples are, for instance, a gas under low pressure or a dilute solution of a solute in some solvent. In many other instances (unfortunately in many that pertain to practical solvent extraction), such an approximation is far from being valid, and quite incorrect estimates of properties of the real systems can result from ignoring the deviations from the ideal.

2.4.1 Ideal Mixtures and Solutions

2.4.1.1 One Liquid Phase

Consider two liquid substances that are rather similar, such as benzene and toluene or water and ethylene glycol. When n_A moles of the one are mixed with n_B moles of the other, the composition of the liquid mixture is given by specification of the mole fraction of one of them [e.g., x_A , according to Eq. (2.2)]. The energy or heat of the mutual interactions between the molecules of the components is similar to that of their self interactions, because of the similarity of the two liquids, and the molecules of A and B are distributed completely randomly in the mixture. In such mixtures, the *entropy of mixing*, which is a measure of the change in the molecular disorder of the system caused by the process of mixing the specified quantities of A and B, attains its maximal value:

$$\Delta_M S_{AB} = -R[x_A \ln x_A + x_B \ln x_B] \quad (2.14)$$

per mole of mixture. This is the molar entropy of mixing expected for an *ideal mixture*. The molar heat of mixing of such a mixture, $\Delta_M H_{AB}$, is zero, since no net change in the energies of interaction takes place on mixing. Therefore, the *molar Gibbs energy of mixing*, in the process that produces an ideal mixture, is:

$$\Delta_M G_{AB} = \Delta_M H_{AB} - T\Delta_M S_{AB} = RT[x_A \ln x_A + x_B \ln x_B] \quad (2.15)$$

Since the mole fractions are quantities that are smaller than unity, their logarithms are negative, $\Delta_M S_{AB}$ is positive, and $\Delta_M G_{AB}$ is negative, the process of mixing being a spontaneous process under the usual conditions. Equations (2.14) and (2.15) suffice to define an ideal (liquid) mixture.

The solute and the solvent are not distinguished normally in such ideal mixtures, which are sometimes called *symmetric ideal mixtures*. There are, however, situations where such a distinction between the solute and the solvent is reasonable, as when one component, say, B, is a gas, a liquid, or a solid of limited solubility in the liquid component A, or if only mixtures very dilute in B are considered ($x_B \ll 0.5$). Such cases

represent *ideal dilute solutions*.

The molar Gibbs energy G of an ideal mixture, whether symmetric or dilute, consists of the molar Gibbs energies of the pure components A and B, weighted by their mole fractions, plus the molar Gibbs energy of mixing, $\Delta_M G_{AB}$.

$$G = x_A G_A^* + x_B G_B^* + \Delta_M G_{AB} \quad (2.16)$$

The *chemical potential*, μ , of a component of the mixture is the partial derivative of the Gibbs energy of the mixture with respect to the number of moles of this component present, the number of moles of all the other components being held constant, as are also the temperature and the pressure. For the component A in a mixture containing also B, C, ..., the chemical potential is $\mu_A = (\partial G / \partial n_A)_{P, T, n_B, n_C, \dots}$, whether the mixture is ideal or not. In the ideal mixture the chemical potential of A is thus obtained from Eqs. (2.15) and (2.16) on carrying out the partial differentiation, yielding:

$$\mu_A = \mu_A^\circ + RT \ln x_A \quad (2.17)$$

As the mole fraction of A in the mixture increases toward unity, the second term in Eq. (2.17) tends toward zero, and the chemical potential of A tends toward the *standard chemical potential*, $\mu_A^\circ = G_A^*$, the molar Gibbs energy (or chemical potential) of A in the realizable *standard state* of pure A, in the sense that pure liquid A is a known chemical substance. A similar equation holds for the component B.

Consider a dilute ideal solution of the solute B (which could be gaseous, liquid, or solid at the temperature in question) in the solvent A. Suppose that more concentrated solutions do not behave ideally and, in particular, the state of pure liquid B cannot be attained by going to more and more concentrated solutions (e.g., by removing A by volatilization). It is possible to define a standard chemical potential pertaining to a hypothetical standard state of the ideal infinitely dilute solution as the limit:

$$\mu_B^\infty = \lim_{x_B \rightarrow 0} (\mu_B - RT \ln x_B) \quad (2.18)$$

Although μ_B tends to $-\infty$ as x_B tends to 0 (and $\ln x_B$ also tends to $-\infty$), the difference on the right-hand side of Eq. (2.18) tends to the finite quantity μ_B^∞ , the standard chemical potential of B. At infinite dilution (practically, at high dilution) of B in the solvent A, particles (molecules, ions) of B have in their surroundings only molecules of A, but not other particles of B, with which to interact. Their surroundings are thus a constant environment of A, independent of the actual concentration of B or of the eventual presence of other solutes, C, D, all at high dilution. The standard chemical potential of the solute in an ideal dilute solution thus describes the solute-solvent interactions exclusively.

A solution in which the solvent A obeys the relationship of Eq. (2.17) and the solute B

obeys the expression

$$\mu_B = \mu_B^\circ + RT \ln x_B \quad (2.19)$$

over a certain (low) concentration range of B in A is said to obey *Raoult's law* for the solvent and *Henry's law* for the solute in this concentration range (Fig. 2.4).

The vapor pressure of the solvent in such a case is given by

$$p_A = x_A p_A^* \quad (2.20)$$

where p_A^* is the vapor pressure of pure liquid A at the given temperature. Conversely, when Eq. (2.20) is followed by the vapor pressure of the solvent, Raoult's law is said to be valid and Eq. (2.17) is obeyed. The vapor pressure of the solute is also proportional to its mole fraction, if Henry's law is obeyed. However, the proportionality constant is not

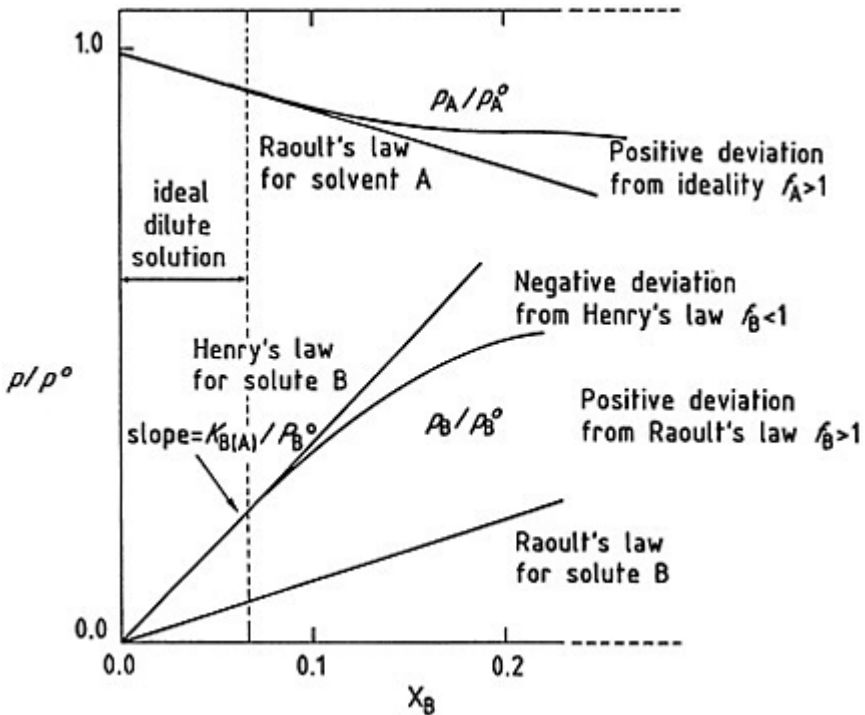


Fig. 2.4 The vapor pressure diagram of a dilute solution of the solute B in the solvent A. The region of ideal dilute solutions, where Raoult's and Henry's laws are obeyed by the solvent and solute, respectively, is indicated. Deviations from the ideal at higher concentrations of the solute are shown. (From Ref. 3.)

$$p_B = K_{B(A)} x_B \quad (2.21)$$

its vapor pressure in the pure liquid state, which may not be attainable at the given temperature. Instead, the expression is followed, where $K_{B(A)}$ is called the *Henry's law constant*, and depends on the chemical natures of both B and A, as the molecular description of the foregoing solutions requires. These considerations apply also when there are several solutes present at low concentrations, so that Raoult's law may apply to the solvent and Henry's law to each of the solutes.

2.4.1.2 Two Liquid Phases

Consider now two practically immiscible solvents that form two phases, designated by ' and ". Let the solute B form a dilute ideal solution in each, so that Eq. (2.19) applies in each phase. When these two liquid phases are brought into contact, the concentrations (mole fractions) of the solute adjust by mass transfer between the phases until equilibrium is established and the chemical potential of the solute is the same in the two phases:

$$\mu_{B'} = \mu_{B''}^{\infty} + RT \ln x_{B''} = \mu_{B'} = \mu_{B'}^{\infty} + RT \ln x_{B'} \quad (2.22)$$

(It is the *difference in the chemical potentials* of the solute that is the driving force for the mass transfer.) This equation can be rewritten in the form:

$$x_{B'} / x_{B''} = \exp[(\mu_{B''}^{\infty} - \mu_{B'}^{\infty}) / RT] = P_B \quad (2.23)$$

where $x_{B'}/x_{B''} = D_B$ is the *distribution ratio* of the solute B (on the mole fraction scale) and the exponential expression is a constant (at a given temperature), called the *distribution constant* of B,* P_B . Equation (2.23) is an expression of *Nernst's distribution law* that states:

The distribution ratio of a solute between two liquid phases at equilibrium is a constant, provided that the solute forms a dilute ideal solution in each phase.

*There is no internationally recommended symbol for the distribution constant, but the symbol P (for partition constant) is common usage in this context. Coordination (complex) chemists (as in Chapter 3) prefer the symbol K_D (for the German "Konstante" for "constant"); sometimes other symbols are used. The important point is that the symbol should be properly defined in the text.

2.4.2 Nonideal Mixtures and Solutions

For most of the situations encountered in solvent extraction the gas phase above the two liquid phases is mainly air and the partial (vapor) pressures of the liquids present are low, so that the system is at atmospheric pressure. Under such conditions, the gas phase is practically ideal, and the vapor pressures represent the *activities* of the corresponding substances in the gas phase (also called their *fugacities*). Equilibrium between two or more phases means that there is no net transfer of material between them, although there still is a dynamic exchange (cf. Chapter 3). This state is achieved when the chemical potential μ as

well as the activity a of each substance are the same in all the phases. It is the inequality of the activities of a substance in two phases that causes some of the substance to transfer from the one (where it is higher) to the other phase, until equality is achieved. The activity of a pure liquid or solid substance is defined as unity. In any mixture, whether ideal or not, the activity of a component A, a_A , is related to its chemical potential by [3]:

$$\mu_A = \mu_A^\circ + RT \ln a_A \quad (2.24)$$

Therefore, if component A is the solvent in a solution, and it obeys Raoult's law when the solution is dilute in all solutes, then $a_A = x_A$ under these conditions. Otherwise, the solution is nonideal, and the deviation from the ideal is described by means of the *activity coefficient*

$$f_A = a_A / x_A \quad (2.25)$$

where a_A is given by Eq. (2.24) or can also be expressed by the ratio of the vapor pressures, p_A/p_A° (neglecting interactions in the vapor phase, otherwise the fugacities have to be employed). As the solution is made more dilute and approaches the dilute ideal solution and the pure solvent, the activity coefficient f_A approaches unity:

$$\lim_{x_A \rightarrow 1} f_A = 1 \quad (2.26)$$

Equation (2.24) can now be rewritten as

$$\mu_A = \mu_A^\circ + RT \ln x_A + \mu_A^E = \mu_A^\circ + RT \ln x_A + RT \ln f_A \quad (2.27)$$

where μ_A^E is the *excess chemical potential* of A in the solution and is equal to $RT \ln f_A$, expressing the deviation of the solution from ideality.

As already mentioned, in solutions that are not symmetric mixtures it is conventional to treat the solute B (or solutes B, C, ...) differently from the treatment accorded to the

solvent A. The activity coefficient is still defined as the ratio a_B/x_B , but the limit at which it equals unity is taken as the infinite dilute solution of B in A:

$$\lim_{x_B \rightarrow 0} f_B = 1 \quad (2.28)$$

so that

$$\mu_B = \mu_B^\infty + RT \ln x_B + \mu_B^E = \mu_B^\infty + RT \ln x_B + RT \ln f_B \quad (2.29)$$

The activity coefficients f_A and f_B may be smaller or larger than unity; hence, the excess chemical potentials μ_A^E and μ_B^E may be negative or positive. Depending on the sign of μ_A^E the solution is said to exhibit positive or negative deviations from Raoult's law. Similarly, positive or negative deviations from Henry's law can be noted (see Fig. 2.4).

The considerations that lead to Eq. (2.22) also apply to the case of non-ideal solutions, except that the terms $RT \ln f_B$ and $RT \ln f_{B'}$ must be added to the left- and right-hand sides. When rearranged into the form of Eq. (2.23), the result is

$$D_B = x_B/x_{B'} = P_B(f_{B'}/f_B) \quad (2.30)$$

Therefore, the distribution ratio of B remains constant only if the ratio of the activity coefficients is independent of the total concentration of B in the system, which holds approximately in dilute solutions. Thus, although solutions of metal chelates in water or nonpolar organic solvents may be quite nonideal, Nernst's law may still be practically obeyed for them if their concentrations are very low ($x_{\text{chelate}} < 10^{-4}$). Deviations from Nernst's law (constant D_B) will in general take place in moderately concentrated solutions, which are of particular importance for industrial solvent extraction (see Chapter 12).

When no distinction between the solvent and the solute in a liquid mixture is made, then nonideal mixtures can still be described by means of an expression similar to Eq. (2.16), but with the addition of a term that is the *excess molar Gibbs energy of mixing*, ΔG_{AB}^E (see also section 2.2). Thus the Gibbs energy per mole of mixture is:

$$G = x_A G_A^* + x_B G_B^* + RT[x_A \ln x_A + x_B \ln x_B] + \Delta G_{AB}^E \quad (2.31)$$

where $\Delta G_{AB}^E = x_A \mu_A^E + x_B \mu_B^E = RT[x_A \ln f_A + x_B \ln f_B]$ is a function of the composition. It is this quantity that has been used in Eq. (2.5) to define the variable $b_{AB}(T)$, which describes the tendency of the mixture to separate into two liquid phases. The excess Gibbs energy is positive ($>2RT$) for mixtures that tend to separate, but is negative when the components have a strong mutual attraction of their molecules. For some mixtures, ΔG_{AB}^E is positive over a part of the composition range and negative over another part.

When a partial derivative of ΔG_{AB}^E relative to the number of moles of a component, say, A, is taken, the result is $RT \ln \gamma_A$. When the partial derivative relative to the temperature is taken, the negative of the excess entropy of mixing, $-\Delta S_{AB}^E$ results. For a regular solution (see section 2.2) this quantity is zero, so that the (excess) heat (enthalpy) of mixing is $\Delta H_{AB}^E = \Delta G_{AB}^E$. The qualifier “excess” for the enthalpy of mixing is put in parentheses, since ideal mixtures have no enthalpies of mixing, so that if heat is evolved or absorbed when two liquids are mixed, the mixture is not ideal and the observed heat (enthalpy) of mixing is always an excess quantity. The enthalpy of mixing, ΔH_{AB}^E may or may not be temperature dependent itself, and may be positive, negative, or may change sign within the composition range. A negative ΔH_{AB}^E is, again, indicative of strong mutual interactions of the molecules of the components.

An example of such behavior is met with in aqueous ethanol (EtOH) at, say, 75°C. At $x_{\text{EtOH}} < 0.24$, the heat of mixing is negative (heat is evolved on mixing the components), but at higher ethanol contents it is positive (heat is absorbed and the mixture cools). In the equimolar mixture $\Delta H_{AB}^E = 220$ but $\Delta G_{AB}^E = 960 \text{ J mol}^{-1}$, due to a negative entropy of mixing, but since $2x_{\text{EtOH}}x_{\text{water}}RT = 1446 \text{ J mol}^{-1}$ for this mixture, a single liquid phase is stable. However, for aqueous 1-butanol (BuOH) at 25°C, $\Delta G_{AB}^E \geq 2x_{\text{BuOH}}x_{\text{water}}RT = 1220 \text{ J mol}^{-1}$ for the near equimolar mixture, although $\Delta H_{AB}^E = 500 \text{ J mol}^{-1}$ only, so the mixture is on the verge of splitting into two liquid phases (it splits when $x_{\text{BuOH}} = 0.485$).

2.4.3 Scales of Concentration

The composition of a mixture need not be given in terms of the mole fractions of its components. Other scales of concentration are frequently used, in particular, when one of the components, say, A, can be designated as the solvent and the other (or others), B, (C, ...) as the solute (or solutes). When the solute is an electrolyte capable of dissociation into ions (but not only for such cases), the *molal scale* is often employed. Here, the composition is stated in terms of the number of moles of the solute, m , per unit mass (1 kg) of the solvent. The symbol m is used to represent the molal scale (e.g., $5 \text{ m} = 5 \text{ mol solute/1 kg solvent}$). The conversion between the molal and the *rational scale* (i.e., the mole fraction scale, which is related to ratios of numbers of moles [see Eq. (2.2)] proceeds according to Eqs. (2.32a) or (2.32b) (cf. Fig. 2.4):

$$m_B = x_B / M_A (1 - x_B) \approx x_B / M_A \quad (2.32a)$$

$$x_B = m_B / (m_B + 1/M_A) \approx m_B M_A \quad (2.32b)$$

where M_A is the molar mass of the solvent A, expressed in kilograms per mole (kg mol^{-1}). For aqueous solutions, for instance, $M_A = 0.018 \text{ kg mol}^{-1}$ and $1/M_A = 55.5 \text{ mol kg}^{-1}$. The approximate equalities on the right-hand sides of Eqs. (2.32a) and (2.32b)

pertain to very dilute solutions. Activity coefficients on the molal scale are designated by γ , so that Eq. (2.29) becomes

$$\mu_B = \mu_{B(m)}^{\infty} + RT \ln m_B + RT \ln \gamma_B \quad (2.33)$$

where, again, $\mu_{B(m)}^{\infty} = \lim_{(m_B \rightarrow 0)} (\mu_B - RT \ln m_B)$ is the standard chemical potential of B on the molal scale, and $\lim_{(m_B \rightarrow 0)} \gamma_B = 1$.

Another widely used concentration scale is the *molar scale*, that describes the number of moles of the solute, c , per unit volume of the solution [i.e., per 1 dm³=1 L (liter)]. The symbol M=mol/L is used to represent this scale [e.g., 0.5 M=0.5 mol solute/1 liter (L) of solution]. Conversion between the molar and molal scales is made according to Eqs. (2.34a) and (2.34b) by means of the density of the solution, ρ , in kg L⁻¹(=g mL⁻¹=g cm⁻³; see Fig. 2.5):

$$c_B = m_B \rho / (1 + m_B M_B) \approx m_B \rho_A / (1 + m_B M_B) \quad (2.34a)$$

$$m_B = c_B / (\rho - c_B M_B) = C_B / (\rho_A - c_B M_B) \quad (2.34b)$$

where M_B is the molar mass of the solute (in kg mol⁻¹). The approximate equalities on the right-hand sides of Eqs. (2.34a) and (2.34b) pertain to very dilute

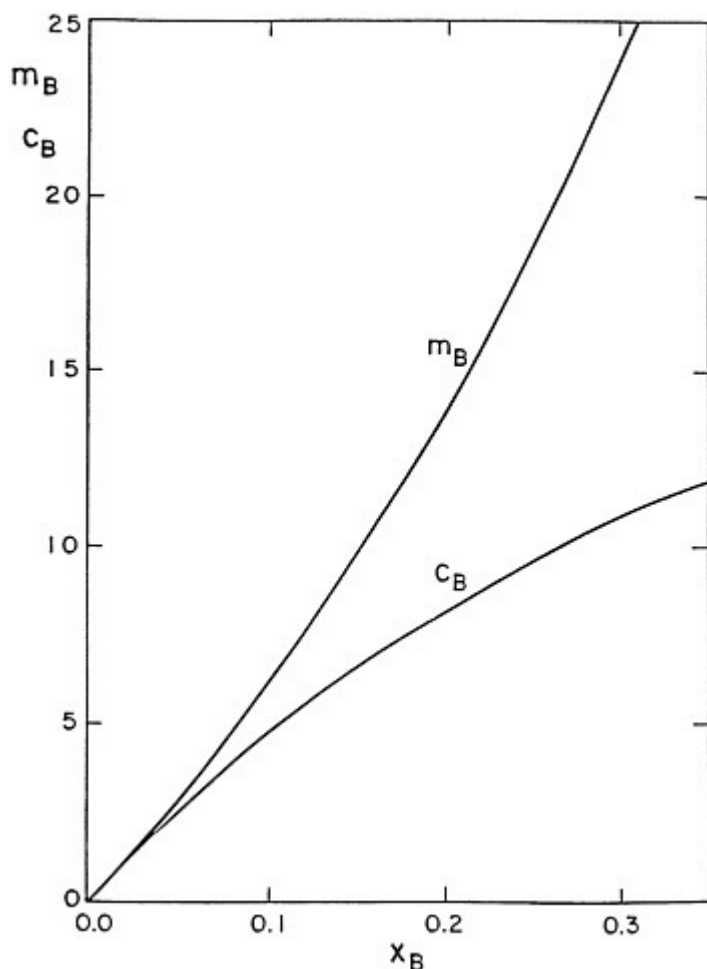


Fig. 2.5 Concentration scales. The molality m_B [in mol/(kg solvent)] and molarity c_B [in mol (L solution) $^{-1}$] of solute B, which has a molar mass M_B 0.100 kg mol $^{-1}$ and a molar volume V_B =0.050 L mol $^{-1}$, in the solvent A, which has M_A =0.018 kg mol $^{-1}$ and V_A =0.018 L mol $^{-1}$ (water), are shown as a function of the mole fraction x_B of the solute. Note that the molarity tends toward a maximal value ($1/V_B$), whereas the molality tends toward infinity as x_B increases toward unity.

solutions. For aqueous solutions at 25°C, ρ_A =0.997 kg L $^{-1}$, and a generally negligible error is made if ρ_A is replaced by 1 for very dilute aqueous solutions. The activity coefficient on the molar scale is designated by y , so that Eq. (2.29) becomes

$$\mu_B = \mu_{B(c)}^\infty + RT \ln c_B + RT \ln y_B \quad (2.35)$$

where, as before, $\mu_{B(c)}^\infty = \lim_{(c_B \rightarrow 0)} (\mu_B - RT \ln c_B)$ is the standard chemical potential of B on the molar scale, $\lim_{(c_B \rightarrow 0)} y_B = 1$.

As an example for such concentration scale conversions, consider 20 wt% tri-*n*-butyl phosphate (TBP, B) in toluene (A). The mole fraction of the TBP is $x_B = (20/M_B)/[(20/M_B) + (80/M_A)] = 0.080$, its molality is $m_B = (0.200/M_B)/0.800 = 0.94 \text{ m}$, and its molarity is $c_B = 0.94 \rho_A / (1 + 0.94 M_B) = 0.65 \text{ M}$ (M_B is taken in kg mol^{-1}).

The quantity μ_B is independent of the concentration scale used, being a true property of the solution, but the three standard chemical potentials $\mu_B^\infty(x)$, $\mu_B^\infty(m)$, and $\mu_B^\infty(c)$ are not equal. Consequently, differences between the standard chemical potentials of a solute in the two liquid phases employed in solvent extraction also depend on the concentration scale used. Thus, P_B defined in Eq. (2.23) is specific for the rational concentration scale, and does not equal corresponding quantities pertaining to the other scales. Therefore, Eq. (2.30) might be rewritten with a subscript (*x*), to designate the rational scale, [i.e., with $D_{B(x)}$ and $P_{B(x)}$]. Similar expressions would then be

$$D_{B(m)} = m_B / m_{B''} = P_{B(m)} (\gamma_B'' / \gamma_B') \quad (2.36a)$$

for the molal scale, and

$$D_{B(c)} = c_B / c_{B''} = P_{B(c)} (\gamma_B'' / \gamma_B') \quad (2.36b)$$

for the molar scale. The subscripts are often omitted, if the context unequivocally defines the concentration scale employed.

2.5 AQUEOUS ELECTROLYTE SOLUTIONS

In many practical solvent extraction systems, one of the two liquids between which the solute distributes is an aqueous solution that contains one or more electrolytes. The distributing solute itself may be an electrolyte. An *electrolyte* is a substance that is capable of *ionic dissociation*, and does dissociate at least partly to ions in solution. These ions are likely to be solvated by the solvent (or, in water, to be hydrated) [5]. In addition to ion-solvent interactions, the ions will also interact with one another: repulsively, if of the same charge sign, attractively, if of the opposite sign. However, ion-ion interactions may be negligible if the solution is extremely dilute. The electrolyte $C_{\nu_+ \nu_-}$ is made up of ν_+ positive ions, or *cations*, C^{z+} , and ν_- negative ions, or *anions*, A^{z-} , or of altogether $\nu = \nu_+ + \nu_-$ ions, and is designated in the following by the symbol CA. A general principle that is always obeyed in electrolyte solutions is that of the electroneutrality of the solution as a whole. This means that the number of cations weighted by their charge numbers z_+

(the number of protonic charges each has) equals that of the anions similarly weighted with z_- . Per unit volume of the solution this can be written as:

$$\sum c_+ z_+ + \sum c_- z_- = 0 \quad (2.37)$$

(Note that z_- is a negative number, so that $v_+ z_+ + v_- z_- = 0$.) The electroneutrality principle is equivalent to stating that it is impossible to produce a solution that contains, for example, only cations or an excess of positive charge (i.e., that ions cannot be considered as independent components of solutions). Only entire electrolytes are components that can be added to a solution.

It is expedient to consider the mean ionic properties of the electrolytes. An electrolyte that dissociates into v ions has a mean molality $m_{\pm} = (v_+ m_+ + v_- m_-)/v$ and a mean molarity $c_{\pm} = (v_+ c_+ + v_- c_-)/v$.

A useful concept that is used when the activities of electrolytes are calculated is that of the *ionic strength* of the solution. This is defined (on the molar scale) as:

$$I = \frac{1}{2} \sum c_i z_i^2 \quad (2.38)$$

where the summation extends over all the cations and anions present in the solution. Since the charges appear to the second power in the expression for the ionic strength, all the terms are positive. An electrolyte that is completely dissociated to univalent ions is designated as a 1:1 electrolyte, examples being aqueous HCl and LiNO₃. If only a single 1:1 electrolyte is present at a molar concentration c_{CA} , then $I = c_{CA}$. According to this notation, MgCl₂, and (NH₄)₂SO₄ are 2:1 and 1:2 electrolytes, respectively, for which $I = 3 c_{CA}$. For 1:1, 2:2, ... (i.e., symmetric) electrolytes, $m_{\pm} = m_{CA}$, and $I = c_{CA}$, $4 c_{CA}$, ... The relationship between m_{CA} and c_{CA} is given by Eq. (2.34.)

2.5.1 Electrolyte Activity*

The chemical potential of an electrolyte may formally (but not experimentally) be considered as being made up from contributions of the individual kinds of ions:

$$\mu_{CA} = v\mu_{\pm} = v_+\mu_+ + v_-\mu_- \quad (2.39)$$

Mean ionic activities, molalities (or molarities), and activity coefficients, designated with subscript $\pm$, are used in equations corresponding to Eqs. (2.33) or (2.35) for electrolyte solutions:

$$\mu_{CA} = v\mu_{\pm}^{\infty} + vRT \ln m_{\pm} + vRT \ln \gamma_{\pm} \quad (2.40)$$

*See also Chapter 6.

Electrolyte solutions are typical nonsymmetrical solutions, in that the solvent is treated according to Eq. (2.27) and the solute according to Eq. (2.40).

The mole fraction concentration scale is generally used for the solvent water, designated by subscript w and having molar mass M_w .

The activity of water is related formally to the molality of the electrolyte by means of the *osmotic coefficient*, ϕ , of the solution:

$$\ln a_w = \ln(p_w / p_w^*) = -\nu m_{\pm} M_w \phi \quad (2.41)$$

As the solution becomes more dilute in the electrolyte (or, in all electrolytes present, where $\nu m_{\pm}$ is replaced by $\sum \nu_i m_{\pm i}$), $\ln a_w$ approaches zero (a_w approaches unity), and ϕ approaches unity.

A definite relationship exists between $\gamma_{\pm}$ and ϕ (the *Gibbs-Duhem relationship*), which may be expressed by the excess Gibbs energy of the solution of an electrolyte:

$$\Delta G^E = \nu m_{\pm} R T [1 - \phi + \ln \gamma_{\pm}] \quad (2.42)$$

This expression is analogous to Eq. (2.3), in that $(1 - \phi)$ expresses the contribution of the solvent and $\ln \gamma_{\pm}$ that of the electrolyte to the excess Gibbs energy of the solution. The calculation of the mean ionic activity coefficient of an electrolyte in solution is required for its activity and the effects of the latter in solvent extraction systems to be estimated. The osmotic coefficient or the activity of the water is also an important quantity related to the ability of the solution to dissolve other electrolytes and nonelectrolytes.

Electrostatic and statistical mechanics theories were used by Debye and Hückel to deduce an expression for the mean ionic activity (and osmotic) coefficient of a dilute electrolyte solution. Empirical extensions have subsequently been applied to the Debye-Hückel approximation so that the expression remains approximately valid up to molal concentrations of 0.5 m (actually, to ionic strengths of about 0.5 mol L⁻¹). The expression that is often used for a solution of a single aqueous 1:1, 2:1, or 1:2 electrolyte is

$$\log \gamma_{\pm} = -A |z_+ z_-| I^{1/2} / [1 + 1.5 I^{1/2}] + b I \quad (2.43)$$

A is a constant that depends on the temperature, and at 25°C $A=0.509$, and b has the value 0.2. (Note that although I is given on the molar scale, the $\gamma_{\pm}$ obtained from the semi-empirical Eq. (2.43) is on the molal scale.) However, Eq. (2.43) does not seem to be valid for 2:2, 3:1, or higher electrolytes (i.e., those with highly charged ions) at practical concentrations. A much more complicated expression results from the Debye-Hückel theory for the osmotic coefficient of the solution, but in the range of validity of Eq.

(2.43), ϕ differs from unity by no more than about -5% . Other modifications of the theory have been suggested to extend its validity beyond the $I=0.5 \text{ mol L}^{-1}$ of Eq. (2.43), but their consideration [3,8] is outside the scope of this book. If I for the solution is $>0.5 \text{ mol L}^{-1}$, and for higher-charge-type electrolytes (2:2, 3:1, etc.), experimental values of $\gamma_{\pm}$ and ϕ can be obtained from suitable compilations [8].

2.5.2 Mixtures Containing Electrolytes

It is important to be able to estimate the values of $\gamma_{\pm}$ for aqueous electrolytes in their mixtures, say B and C. An expression that has been found to be valid under wide conditions is *Harned's rule*:

$$\log \gamma_{\pm B} = \log \gamma_{\pm B}^0(I_m) - \alpha_{B(C)} I_{mC} \quad (2.44a)$$

and

$$\log \gamma_{\pm C} = \log \gamma_{\pm C}^0(I_m) - \alpha_{C(B)} I_{mB} \quad (2.44b)$$

where in Eq. (2.44a) the activity coefficient of electrolyte B in its mixture with C is $\gamma_{\pm B}$, the latter electrolyte contributing I_{mC} to the total ionic strength on the molal scale, $I_m = \frac{1}{2} \sum m_i z_i^2$, and $\gamma_{\pm B}^0(I_m)$ is the activity coefficient of B when present alone at molal ionic strength I_m . A similar expression, Eq. (2.44b), holds for $\log \gamma_{\pm C}$ of electrolyte C in the mixture.

Consider, for example an aqueous solution that is 1 m in HCl (electrolyte B) and 2 m in LiCl (electrolyte C), at a total ionic strength on the molal scale of $I_m=3 \text{ m}$. The activity coefficient of HCl in 3 m HCl is 1.316 [its logarithm is $\log \gamma_{\pm \text{HCl}}^0(I_m)=0.119$] and $\alpha_{\text{HCl}(\text{LiCl})}=0.004$ at this total molality. Hence, $\log \gamma_{\pm \text{HCl}}=0.119-2 \times 0.004=0.111$ or $\gamma_{\pm \text{HCl}}=1.291$ in this mixture. Similarly, $\gamma_{\pm \text{LiCl}}^0(I_m)=1.174$ ($\log \gamma_{\pm \text{LiCl}}^0(I_m)=0.070$) for 3 m LiCl and $\alpha_{\text{LiCl}(\text{HCl})}=-0.013$, so that $\log \gamma_{\pm \text{LiCl}}=0.070-1 \times (-0.013)=0.083$ or $\gamma_{\pm \text{LiCl}}=1.211$ in this mixture. Thus, the activity coefficient of HCl is lower and that of LiCl is higher in the mixture than in the single electrolyte solutions at the same total molality. If the HCl is present at trace concentration in 3 m LiCl, its activity coefficient would be somewhat reduced to $\gamma_{\pm \text{HCl}, \text{trace}}=1.279$, whereas that of trace LiCl in 3 m HCl would be somewhat increased to $\gamma_{\pm \text{LiCl}, \text{trace}}=1.285$.

The values of $\alpha_{B(C)}$ and $-\alpha_{C(B)}$ are interrelated. For very simple systems they are equal, but generally they are not and must be obtained from the literature [9]. They also depend on the total ionic strength, I , but not on the composition relative to the two electrolytes. It must also be noted that Eq. (2.44) is a semi-empirical expression, and many systems do not obey it, requiring added terms in I_{mC}^2 or I_{mB}^2 .

When one electrolyte, say, B, is present at very low concentrations in the presence of the other, say, C, then $I_{mC} \approx I_m$, and if this ionic strength is fixed, then according to Eq. (2.44) the activity coefficient $\gamma_{\pm B}$ of the trace electrolyte B becomes independent of its own concentration in the mixture. That is, $\log \gamma_{\pm B} = \log \gamma_{\pm B}^\circ(I_m) - \alpha_{B(C)} I_m \neq f(m_B)$. The same is true if there are several electrolytes present, all at concentrations much lower than that of C, which is fixed.

This is the basis of the *ionic medium method*, where an electrolyte C, such as sodium perchlorate, is kept in the solution at a fixed and high concentration (e.g., 3 mol L⁻¹). This practice permits the concentrations of reactive electrolytes, that is, those that provide ions that participate in reactions (contrary to the “inert” ions of the medium electrolyte) to be varied below certain limits at will, without changes in their activity coefficients. These are determined entirely by the natures of the electrolytes and the fixed concentration of C. Being constant quantities under these conditions, the activity coefficients can be incorporated into *conditional equilibrium constants* for the reactions where the ions participate. (It becomes also immaterial for the use of the constant ionic medium whether the ionic strength is specified in terms of a constant molality or molarity, although the numerical values of the equilibrium constants depend on this specification.)

The nonideality of electrolyte solutions, caused ultimately by the electrical fields of the ions present, extends also to any nonelectrolyte that may be present in the aqueous solution. The nonelectrolyte may be a co-solvent that may be added to affect the properties of the solution (e.g., lower the relative permittivity, ϵ , or increase the solubility of other nonelectrolytes). For example, ethanol may be added to the aqueous solution to increase the solubility of 8-hydroxyquinoline in it. The nonelectrolyte considered may also be a reagent that does not dissociate into ions, or one where the dissociation is suppressed by the presence of hydrogen ions at a sufficient concentration (low pH; cf. Chapter 3), such as the chelating agent 8-hydroxyquinoline.

In any event, the activity coefficient of the nonelectrolyte, designated by subscript N, generally follows the *Setchenov equation*:

$$\log y_N = \log y_N^\circ + k_{N,CA} c_{CA} \quad (2.45)$$

where y_N° is the activity coefficient of the nonelectrolyte on the molar scale in the absence of the electrolyte CA, and $k_{N,CA}$ is a proportionality constant that depends on the natures of N and the electrolyte CA, but is independent of their concentrations. The larger the molar volume and the lower the polarity of the nonelectrolyte N, generally the larger $k_{N,CA}$ is. The better hydrated the ions of the electrolyte CA are, the larger, again, $k_{N,CA}$ is. Since the values of $k_{N,CA}$ are generally positive, the activity coefficient of the nonelectrolyte in the solution increases in the presence of the electrolyte, and for a given activity of the former (e.g., determined by equilibrium with excess insoluble nonelectrolyte), its concentration must decrease. The nonelectrolyte is then said to be salted-out by the electrolyte. As an illustration, consider the salting-out of benzene by CsCl, LiCl, and BaCl₂; the corresponding values of $k_{N,CA}$ are 0.088, 0.141, and 0.334. These values are ordered according to the strength of the hydration of these salts (see Table 2.4). A poorly hydrated salt, such as tetramethylammonium bromide even salts-in

benzene, with $k_{N,CA} = -0.24$. Equation (2.45) may be obeyed up to electrolyte concentrations of several moles per liter. The contributions of several electrolytes are, within limits, additive.

2.6 ORGANIC SOLUTIONS

Solutions in organic solvents or in mixed aqueous-organic solvents, on the whole, behave not very differently from purely aqueous solutions. In particular, if the relative permittivity of the solvent (see Table 2.1 for values) is roughly $\epsilon > 40$, electrolytes are, more or less, completely dissociated into ions. On the other hand, if $\epsilon < 10$, only slight or practically no ionic dissociation takes place. Furthermore, solutions in organic or mixed aqueous-organic solvents lack the three-dimensional cooperative hydrogen bond network that characterizes aqueous solutions. Many “anomalous” properties of aqueous solutions that depend on this structured nature of water become normal properties in organic solvents.

2.6.1 Electrolyte Transfer

The presence of an organic solvent in an aqueous mixture affects the activity coefficient of an electrolyte relative to its value in water, even when the electrolyte is present at such low concentrations that, in the absence of the co-solvent, effectively $y_{\pm}$ would have been unity. Accordingly, in mixed aqueous-organic solvents there is a *primary medium effect* on the activity coefficient, reflecting the interactions of the ions with their mixed-solvent surroundings, compared with their interactions with a purely aqueous environment. If the content of water in the mixture is continuously reduced until the pure organic component is reached, this effect is expected to increase and reach a value characteristic of the pure solvent. This quantity can be expressed in terms of a standard molar *Gibbs energy of transfer* of the electrolyte CA from water, aq, to a new medium, org (which may also be a nonaqueous solvent that is immiscible with water): $\Delta_i G^\circ(\text{CA}, \text{aq} \rightarrow \text{org})$ (see section 2.3); here, we use the symbols aq and org to indicate the pure liquids. Thus, when an infinitely dilute solution of the electrolyte CA in the new medium is compared with a corresponding solution in water, we have

$$\Delta_i G^\circ(\text{CA}, \text{aq} \rightarrow \text{org}) = \mu_{\text{CA}}^\infty(\text{org}) - \mu_{\text{CA}}^\infty(\text{aq}) \quad (2.46)$$

Note that the μ_{CA}^∞ are the standard molar Gibbs energies of solvation of the (combined ions of the) electrolyte [5]. Alternatively, a transfer activity coefficient can be defined as

$$\ln \gamma_i^0(\text{CA}, \text{aq} \rightarrow \text{org}) = \Delta_i G^\circ(\text{CA}, \text{aq} \rightarrow \text{org}) / RT \quad (2.47)$$

The superscript zero designates the standard state of infinite dilution, where only solute-solvent but no solute-solute interactions take place. As mentioned in

section 2.3, tables of $\Delta_t G^\circ(\text{ion, aq} \rightarrow \text{org})$ [5–7] may be used to obtain the values for ions, which can then be combined to give the value of $\Delta_t G^\circ(\text{CA, aq} \rightarrow \text{org})$, for the desired electrolyte. For example, for the solvent org=nitrobenzene (PhNO_2), $\Delta_t G^\circ(\text{Na}^+, \text{H}_2\text{O} \rightarrow \text{PhNO}_2)=36 \text{ kJ mol}^{-1}$, $\Delta_t G^\circ(\text{Cs}^+, \text{H}_2\text{O} \rightarrow \text{PhNO}_2)=18 \text{ kJ mol}^{-1}$, and $\Delta_t G^\circ(\text{I}_3^-, \text{H}_2\text{O} \rightarrow \text{PhNO}_2)=-23 \text{ kJ mol}^{-1}$. Therefore, $\Delta_t G^\circ=13 \text{ kJ mol}^{-1}$ for NaI_3 and -5 kJ mol^{-1} for CsI_3 transferring from water to nitrobenzene, signifying the preference of the former salt for water (positive $\Delta_t G^\circ$ and of the latter salt for nitrobenzene (negative $\Delta_t G^\circ$).

When nonnegligible concentrations of the electrolyte are present in the organic solvent, ion-ion interactions superimpose on the ion-solvent ones, or *the secondary medium effect*. Although an equation similar to Eq. (2.43) may be used for determining the activity coefficient in the new medium, it is necessary to employ the appropriate value of A in this equation that depends on the relative permittivity of the medium: $A(\text{org})=A(\text{aq})(\epsilon_{\text{aq}}/\epsilon_{\text{org}})^{3/2}$. Unless very water-rich mixed solvents are used, different numerical values of the parameters in the denominator and the second term on the right-hand side of Eq. (2.43) have to be employed.

2.6.2 Ion Pair Formation

A further complication that sets in when organic or mixed aqueous-organic solvents are used, which is aggravated when the relative permittivity of the medium, ϵ , falls below 40, is *ion pairing*. This phenomenon does occur in purely aqueous solutions, mainly with higher-valence-type electrolytes: 2:2 and higher, and with 2:1 or 1:2 electrolytes only at high concentrations. Ion pairs may also form in aqueous solutions of some 1:1 electrolytes, provided the ions are poorly hydrated and can approach each other to within $<0.35 \text{ nm}$. Such ion pairs are of major importance in solvents that are relatively poor in water or that are nonaqueous.

Ion pairing is due to electrostatic forces between ions of opposite charges in a medium of moderate to low relative permittivities. It should be distinguished from complex formation between metal cations and anionic ligands, in which coordinative bonds (donation of an electron pair) takes place. One distinguishing feature is that, contrary to complex formation, the association is nondirectional in space. The association of a cation and an anion to form an ion pair can, however, be represented as an equilibrium reaction by analogy to complex formation with an equilibrium constant K_{ass} [3,5]. If α is the fraction of the electrolyte that is dissociating into ions and therefore $(1-\alpha)$ is the fraction that is associated, then

$$K_{\text{ass}} = (1-\alpha)y_{\text{N}} / c_{\text{CA}}\alpha^2 y_{\pm}^2 \quad (2.48)$$

where y_{N} is the activity coefficient of the undissociated part [treated according to Eq. (2.45) or set, with good approximation, equal to unity] and $y_{\pm}$ is the mean ionic activity coefficient of the dissociated part of the electrolyte. This quantity can be calculated from the extended Debye-Hückel Eq. (2.43), provided that the value of A appropriate for the

medium having the relative permittivity ϵ is used and that the ionic strength I takes into account the incomplete dissociation: ac replaces c in Eq. (2.38).

According to *Bjerrum's theory*, the association constant is proportional to $a \cdot b$ times a certain function of b , $Q(b)$. The quantity a is the ionic distance parameter (i.e., the distance between the centers of the cation and the anion). The quantity b is proportional to the absolute value of the product of the charges of the cation and the anion and inversely proportional to the absolute temperature, T , the relative permittivity (dielectric constant) of the solvent, ϵ , and to the distance a . At 25°C

$$\log b = \log |z_+ z_-| + 1.746 - \log \epsilon - \log a \text{ (in nm)} \quad (2.49)$$

The function $Q(b) = \int_0^b t^{-4} e^{-t} dt$ (t is an auxiliary variable) mentioned earlier can be approximated by e^b/b^4 at high values of b , or be read from a figure [3]. The value of K_{ass} at 25°C is given by

$$\log K_{\text{ass}} \text{ (in L mol}^{-1}\text{)} = 3 \log |z_+ z_-| + 6.120 - 3 \log \epsilon + \log Q(b) \quad (2.50)$$

Finally, values of α can be obtained from K_{ass} by means of Eq. (2.48), provided that $y_{\pm}$ is calculated iteratively [use Eqs. (2.34), (2.38), and (2.43)].

Solvent-separated ion pairs, in which the first solvation shells of both ions remain intact on pairing may be distinguished from solvent-shared ion pairs, where only one solvent molecule separates the cation and the anion, and contact ion pairs, where no solvent separates them (Fig. 2.6). The parameter a reflects the minimum distance by which the oppositely charged ions can approach each other. This equals the sum of the radii of the bare cation and anion plus 2, 1, and 0 diameters of the solvent, respectively, for the three categories of ion pairs. Since a appears in Eq. (2.49), and hence, also in $Q(b)$ it affects the value of the equilibrium constant, K_{ass} . The other important variable that affects K_{ass} is the product $T\epsilon$ and, at a given temperature, the value of the relative permittivity, ϵ . The lower it is, the larger b is and, hence, also K_{ass} .

When the relative permittivity of the organic solvent or solvent mixture is $\epsilon < 10$, then ionic dissociation can generally be entirely neglected, and potential electrolytes behave as if they were nonelectrolytes. This is most clearly demonstrated experimentally by the negligible electrical conductivity of the solution, which is about as small as that of the pure organic solvent. The interactions between solute and solvent in such solutions have been discussed in section 2.3, and the concern here is with solute-solute interactions only. These take place mainly by dipole-dipole interactions, hydrogen bonding, or adduct formation.

2.6.3 Other Kinds of Interactions

Dipole-dipole interactions between polar solute particles in organic solvents may be

either repulsive (“head-head” interactions) or attractive (“head-tail” interactions). The former are rare and are due to steric hindrance around one of the poles of the dipole, which causes the ends of neighboring molecules that carry partial charges of the same sign to approach it (see Fig. 2.1b). The attractive dipole interactions lead to solute aggregation, either to dimers, where the “head” of each of the two partners is near the “tail” of the other (see Fig. 2.1a), or to larger, chainlike or cyclic aggregates (oligomers), for which each member of the chain or cycle interacts with a neighbor at both ends (see Fig. 2.1c). The

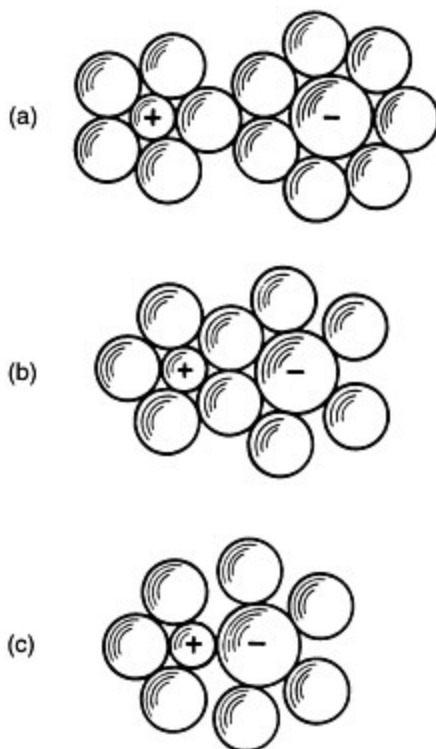


Fig. 2.6 Ion pairs. A two-dimensional representation of (a) a solvent-separated pair of ions, each still retaining its intact shell of solvating solvent molecules; (b) a solvent-sharing ion pair, which has lost some of the solvent between the partners, so that one layer of solvent shared between them separates them; (c) a contact ion pair, the cation and anion being contiguous.

dipole moment, μ , is proportional to the spatial separation of the (partial) charges that cause the particle to be dipolar and is generally considered to “reside” midway between these charges. The energy of dipole-dipole interactions depends on the product of the squares of the dipole moments, μ , of the partners (i.e., on μ^2 , if there is a single kind of interacting species) and inversely on the sixth power of the distance between the centers

of the dipoles. Such an energy is not very large, ranging between 1 and 10 kJ mol⁻¹ (i.e., 0.4 to 4 times the mean thermal energy, RT , that tends to disperse the aggregates). Tertiary amine salts in hydrocarbon solvents are typical examples of such aggregated solutes.

Hydrogen bonding occurring between solute particles also leads to aggregation, when the molecule of the solute contains both a hydrogen atom bonded to an electronegative atom and another electronegative atom that can accept a hydrogen bond. Typical of such solutes are carboxylic acids and acidic phosphates. Cyclic dimers are formed (see Fig. 2.1d), bonded by two hydrogen bridges, the energy of each being about 20 kJ mol⁻¹. Such cyclic dimers are quite stable. If the numbers of ionizable hydrogen atoms and electronegative accepting sites do not match, then noncyclic aggregates, possibly larger than dimers, are formed by hydrogen bonding. One aspect of such an aggregation is a drastic increase in the viscosity of the solution with increasing concentrations.

The mean aggregation number of a solute at molality m , forming various oligomers (i -mers, $i=1, 2, 3, \dots$), is

$$\bar{n} = m / \sum_i m_i = \sum_i i m_i / \sum_i m_i \quad (2.51)$$

the molality of each oligomeric species being weighted by the number of monomers it contains. The equilibrium constants for the aggregation reactions can be obtained from the dependence of $\bar{n}$ on m by the methods discussed elsewhere in this book (see Chapter 3).

The third kind of solute-solute interaction, *donor-acceptor adduct formation*, tends to form 1:1 adducts between the molecules of two different kinds of solute, rather than to lead to self-aggregation. Adduct formation results when one partner has a donor atom (i.e., an atom with an exposed pair of unbonded electrons), such as the nitrogen atom in triethylamine, and the other has an acceptor atom (i.e., one with an orbital that can take up such a pair of electrons), such as antimony in antimony pentachloride or even the hydroxyl hydrogen atom in octanol. Although adduct formation formally can be said to take place in dipole-dipole interactions between two different kinds of molecules, this is rare, since the strong dipoles required naturally participate also in self-aggregation. But if one kind of molecule is a donor only and the other is an acceptor only, as occurs when octanol (the oxygen atom of which has two exposed non-bonded pairs of electrons) in the foregoing example is replaced by antimony pentachloride, then self-aggregation is precluded, and only mutual adduct formation can take place (see following).

Hydrogen bond formation between dissimilar molecules is an example of adduct formation, since the hydrogen atom that is bonded to an electronegative atom, such as oxygen or nitrogen, is a typical acceptor atom. The ability of molecules to donate a hydrogen bond is measured by their Taft-Kamlet *solvatochromic parameter*, α , (or α_m for the monomer of self-associating solutes) (see Table 2.3). This is also a measure of their acidity (in the Lewis sense, see later, or the Brønsted sense, if protic). Acetic acid, for instance, has $\alpha=1.12$, compared with 0.61 for phenol. However, this parameter is not